Persistent and Snap Decision-Making*

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To avoid unfavorable inferences about her ability, an expert might cling to her original opinion and ignore valuable new information in formulating subsequent opinions. Conceivably, the expert might decline an initial opportunity to offer an opinion, delaying the opinion formation until more accurate information has arrived. However, we show that reputational concerns often lead an expert to express an opinion at the first opportunity, thereby making a *snap decision*.

Keywords: reputation, herding, persistence of opinion, snap decision-making

JEL classification: D82, D83, D90

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1. Introduction

It is widely observed that experts often cling to their original opinion and that such consistency is valued by others.[1] For example, when a politician makes a statement, it is often seen as a pledge and it is difficult for her to renege on it without being subject to criticism. Likewise, economists who predict economic trends that are inconsistent with their previous predictions tend not to be trusted. However, remaining consistent could lead an expert to ignore valuable new information in formulating subsequent opinions, which often results in bad outcomes.

Many studies examine *reputation concern* as a force driving consistent behavior (Trueman, 1990, Li, 2007, and Falk and Zimmermann, 2017). In these studies, an expert recommends a choice to an evaluator. The expert has two opportunities to make a recommendation and, before each opportunity, she receives an independent signal conditional on the true state of the world. The accuracy of the signals positively correlates with the expert's ability, and this accuracy increases over time, such that the second signal is more accurate than the first. The expert has reputational concerns about her ability, In this setting, although a recommendation based on the second signal is preferable for the evaluator, the expert has an incentive to cling to the recommendation based on the first. This is because an expert with high abilities would receive highly accurate signals and is less likely to receive inconsistent signals. Hence, the expert behaves consistently to convey the impression that she received consistent signals because she has high abilities.

Although this explanation is intuitive, one may argue that if consistent behavior is inevitable and leads to a less valuable recommendation, the expert would remain silent during the initial opportunity to avoid revealing any inconsistency between the received signals. If this is accurate, unless the expert is forced to make a decision at the

[1] In social psychology, for instance, Cialdini (2006) summarizes and discusses the evidence on consistent behaviors.

initial opportunity, such consistent behavior resulting in less valuable recommendations would not be observed. In the leading example, the politician should be careful not to commit herself in a speech because she is better off waiting for additional information, rather than declaring an intention immediately. However, in reality, people often make hasty decisions. Nutt (2002) discusses how hasty decision-making is a major cause of poor decisions using several case studies of companies. Consider the following case as an example. Shell UK decided to undertake a deep-sea disposal of a spar (a floating oil storage facility). Before the disposal process began, Greenpeace activists began a public campaign opposing it, which damaged Shell UK's public image, and forced it to find another way to dispose of the spar. As Nutt (2002) discusses, the Shell officials did not need to rush in making their decision and could have waited until a public debate about the best disposal method had occurred.

In this study, we seek to answer the question of why experts convey their earlier opinions rather than waiting for new information, even when it is well known that this leads them to seek unproductive consistency of behavior, which, in turn, leads to bad outcomes. In our model, as in previous studies, the expert has two opportunities to make a recommendation. However, this study takes a further step by examining a case where the expert has two options during the first opportunity; that is, the expert can either reveal or conceal the information about the signal that she has received at the initial opportunity. We find a sufficient condition under which the expert has an incentive to reveal the information during the first opportunity, even if it leads to unproductively consistent behavior. Therefore, even when the expert is not forced to make a decision at the initial opportunity, consistent behavior can be observed. We refer to the behavior of breaking silence during the first opportunity and ensuring consistent behavior subsequently as *snap decision-making*.

The intuition is as follows. The expert considers that by remaining silent during

the first period, the evaluator will believe that the expert's final decision is based on the second signal. Conversely, if the expert makes a decision during the first period and employs the strategy of behaving consistently, the evaluator will believe that the expert's decision is based on the first signal. If the expert waits for a more accurate signal and matches her recommendation to the true state, this will be attributed partly to the improvement of the signal and less to the expert's ability than is the case with acting on the first signal. By contrast, if the recommendation does not match the state when the expert acts on the second signal, this is attributed to ability more than is the case with the first signal. Therefore, if the expert remains silent, her ability is discounted regardless of whether her recommendation matches the state. This reveals the advantage of snap decision-making.

By contrast, if the expert remains silent initially, her ultimate recommendation is more likely to match the true state than is the case under snap decision-making. Because the expert's reputation is higher when her recommendation matches the state, this indicates the disadvantage of snap decision-making.

In our model, we find that under symmetric information between the expert and evaluator concerning the expert's ability, if the expert's utility function for the assessment is strictly convex[2] and some additional assumptions are satisfied, an equilibrium with snap decision-making exists, which results in higher expected utility than does remaining silent. Conversely, there is also an equilibrium where the expert reveals no information about the first signal, and makes an ultimate recommendation based only on the second signal.[3] However, making a snap decision results in higher utility than does remaining silent and, thus, there is an incentive to reveal the first signal if this is possible. This implies that the equilibrium revealing no information about the first signal violates a natural equilibrium selection criterion, called the neologism-

[2]Convex utility for the assessment is justified when the whole game is repeated. See Li (2007) for details.

[3]This equilibrium includes a babbling strategy in the first stage.

proofness (Farrell, 1993). In this sense, the snap decision is the unique prediction of our model.

Our study suggests that, focusing on the neologism-proof equilibria, even when the expert can remain silent, she may decide to engage in unnecessary snap decision-making, leading to a poor decision. To avoid such unproductive recommendations, our study suggests that the client should not only refrain from listening to earlier opinions but also silence the expert during the first period by imposing a cost. For example, in the context of decision-making in organizations, creating a norm that encourages deliberation may be a solution. In political contexts, electorates should criticize politicians if they comment on a future political decision without waiting for new information. Conversely, an intentionally noncommittal (and thus ambiguous) statement should be praised.

The remainder of the paper is organized as follows. The next section reviews the related literature. In section 3, we explain our model. Section 4 addresses the case where the expert is forced to recommend a choice in each period and provides a sufficient condition under which the expert behaves consistently. In section 5, we analyze the case in which the expert can remain silent in the first period. Section 6 includes several extensions for analyzing the consistent behavior, and section 7 draws conclusions. All proofs appear in the appendices.

2. Relationship to the literature

Reputation concern distorts an expert's decision-making.[4] The relation between reputation concern and persistency in behaviors has been investigated by several studies,

[4]Reputation concern is also interpreted as career concern. In this literature, Holmström (1999) discusses how the career concern of an agent can be detrimental in a dynamic moral hazard model. As in our model, he assumes symmetric information about the agent's ability. In the cheap talk literature, Ottaviani and Sørensen (2006a,b) provide a model with a general signal structure and discuss how reputation concern distorts the incentive for truth-telling.

such as Kanodia, Bushman and Dickhaut (1989), Trueman (1990), Boot (1992), Prendergast and Stole (1996), Levy (2004), Majumdar and Mukand (2004), Ferreira and Rezende (2007), Li (2007), Sabourian and Sibert (2009), and Falk and Zimmermann (2017). Except for Prendergast and Stole (1996), these studies deal with binary state cases, as in our model. Prendergast and Stole (1996) deal with a continuous state case in which the true state and signals are drawn from normal distributions. They show that the reputation-concerned expert becomes conservative over time.[5]

Similar to the studies identified above, we show that the expert behaves consistently. In this respect, Li (2007) and Falk and Zimmermann (2017) (hereafter, FZ) are the closest works to our study; in their models, as in ours, the expert has two opportunities to recommend a choice, and the signal accuracy grows over time. Furthermore, FZ provide experimental evidence. However, in Li and FZ (and, indeed, in most other studies), the focus is on the second-period behavior. The expert is assumed to recommend a choice in the first period.[6] The main innovation of our study is in endogenizing the first-period recommendation (snap decision-making) by showing that it has a reputational advantage compared with remaining silent.

This study is also related to the literature on herding behavior because persistency in the expert's recommendation is seen as herding in relation to her initial recommendation. In a seminal work, Scharfstein and Stein (1990) develop a reputational herding model in which each expert sequentially makes a decision and is concerned about her reputation. These authors show that in each equilibrium, the second expert follows the decision of the first, regardless of the second expert's signal.

Belief persistence, a well-known confirmation bias, is related to our study. This bias

[5] Falk and Zimmermann (2018) provide experimental evidence for this model. Bernhardt, Wan and Xiao (2016) provide a structural model for estimating the size of bias for consistency.

[6] Related to this point, Levy (2004) examines the case in which the expert has a choice not to advise. In another study, Liu and Sanyal (2012) consider a situation in which the second-period recommendation is exogenously given and show that reputation concern distorts the incentive for the first-period recommendation.

states that once a belief has been formed, people are reluctant to change it (Nickerson, 1998). For example, Rabin and Schrag (1999), provide a model that describes how the preference for belief persistence leads to persistency in behavior. In our study, by contrast, the belief is updated under Bayesian assumptions, but the behavior persists. Eyster (2002) and Yariv (2005) also provide models that describe preferences for consistency.

As noted above, we address another type of seemingly irrational behavior, namely, snap decision-making. This is our novel approach and the key contribution of our study. The logic used to explain this behavior can be seen as a kind of *self-handicapping strategy*—that is, taking a lower performance action to maintain one's self-esteem (see Tirole, 2002). In our model, by revealing only the first signal that yields lower performance, the expert can maintain her reputation regarding ability. Persistent behavior enables the expert to commit to a strategy of not revealing the second signal. This point is similar to the work of Ferreira and Rezende (2007), who show that information disclosure works as a commitment device to make behavior persistent. In Ferreira and Rezende (2007), the benefit of persistent behavior is independent of reputation concern, whereas, in our model, the benefit of persistent behavior comes from pure reputation concern. In a different setting, Grubb (2011) considers the case where the expert needs to reveal the signal if she recommends a choice, and shows that remaining silent is an equilibrium.[7] This contrasts with our result.

3. Model

The model comprises two players, an expert (sender) and an evaluator (receiver), and the state of the world, denoted by $\omega \in X = \{x, y\}$. To simplify the discussion, we

[7] Relatedly, in a principal–agent setting, Demski and Sappington (1986) show that an informed principal is reluctant to release her information. In their setting, the motivation is to encourage the agent's costly information acquisition.

assume that each state realization is equally likely. The expert recommends one of two alternatives, x and y. The expert has two opportunities to make a recommendation. In period 1, not making a recommendation is allowed, which is denoted by $\varnothing$. In opportunity $t \in \{1, 2\}$, the expert receives an independent signal $s_t \in S = \{x, y\}$ with accuracy θ_t (i.e., $p(s_t = \omega \mid \omega) = \theta_t$ for each $\omega \in \{x, y\}$). The accuracy of the signal grows in terms of the odds ratio, that is, the odds ratio of the signal received at the second opportunity is $\frac{\theta_2}{1-\theta_2} = (1+\alpha)\frac{\theta_1}{1-\theta_1}$, where $\alpha > 0$.[8] The accuracy growth rate α is common and known to each player. The accuracy at the first opportunity, θ_1, is referred to as the expert's *ability*. This assumption means that given the expert's ability, θ_1, she receives a more accurate signal about the state of the world, on which she will base a recommendation, at the time of her second opportunity.[9]

We assume that θ_1 is distributed according to cumulative density function F on set $D \subseteq [1/2, 1]$ and that its associated probability density function is denoted by f.[10] Assume that the cardinality of D is no less than 2.[11]

The expert does not know her ability, but receives a signal about it, denoted by τ. We call this signal the *ability signal*. On receiving the ability signal, τ, the probability that ability is θ is denoted by $f(\theta \mid \tau)$.[12] Let $T \neq \varnothing$ be denoted by the set of ability signals. We impose no restriction on T and $(f(\cdot \mid \tau))_{\tau \in T}$. Therefore, this includes the known ability case as a special case, such that $T = D$, and for each $\theta \in D$, $f(\theta \mid \theta) = 1$ and $f(\theta' \mid \theta) = 0$ for each $\theta' \neq \theta$.[13] The information structure includes the case in which the expert has no private information about her ability by assuming that $|T| = 1$.

[8] Equivalently, $\theta_2 = \frac{(1+\alpha)\theta_1}{1+\alpha\theta_1}$.

[9] This assumption does not imply that the first signal is more informative about the expert' s ability than the second signal.

[10] If D is a countable set, for each x, $f(x)$ is the probability that the expert's ability $\theta_1 = x$.

[11] The set D is allowed to be discrete or continuous.

[12] If D is a continuum, f is the conditional density function of θ.

[13] If D is a continuum, $f(\cdot \mid \theta)$ is the density of a Dirac measure placing probability 1 on θ.

The true state of the world is revealed to the evaluator after period 2; subsequently, the evaluator updates his belief regarding the expert's ability.

For each $\tau \in T$, let $r(\tau) = (r_1(\cdot : \tau), r_2(\cdot, \cdot, \cdot : \tau))$ denote the interim strategy, where $r_1(s_1 : \tau) : S \to R_1 = \{x, y, \varnothing\}$ and $r_2(r_1, s_1, s_2 : \tau) : R_1 \times S^2 \to R_2 = \{x, y\}$. That is, r_1 is the first recommendation and r_2 is the second. The first recommendation depends on τ and the first signal s_1, whereas the second recommendation depends on the first recommendation r_1 and the second signal s_2. Let $\beta : R_1 \times R_2 \times X \to \Delta(D)$ denote a belief system regarding the expert's ability.

The expert obtains profit $K \geqslant 0$ only when her final recommendation coincides with the true state of the world. The expert does not incur any cost to change her recommendation. More importantly, the expert also profits from the evaluator's assessment of the expert's ability θ_1 (reputation concern). The expert's von Neumann–Morgenstern (vNM) utility is:

$$I(r_2 = \omega)K + \Phi(\mathbb{E}_\beta[\theta_1 \mid r_1, r_2, \omega]), \tag{1}$$

where I is the indicator function $I(E) = 1$ when the event E is true and $I(E) = 0$ otherwise. $\mathbb{E}_\beta[\theta_1 \mid r_1, r_2, \omega]$ is the expectation of θ_1 with the evaluator's updated belief regarding θ_1. Φ is a continuously differentiable increasing function that is sometimes assumed to be convex. Li (2007) provides one justification for this. In her model, the whole game is performed twice. The reputation made in the first game affects the payment for the expert's recommendation in the second. Li shows that the payment is a convex function of the assessment.

Finally, assume that the expert maximizes the expected value of (1). Figure 1 illustrates the timeline of this model.

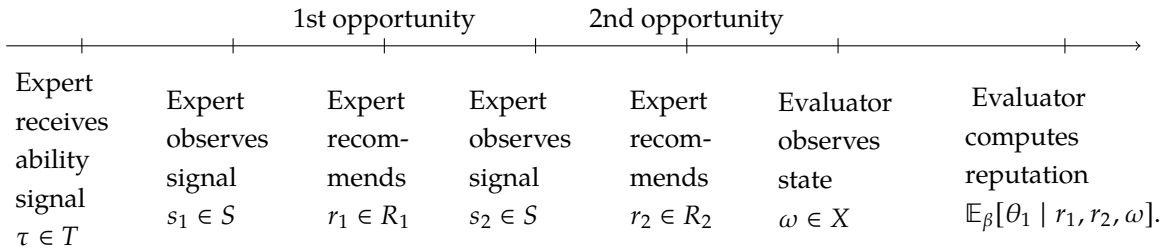

Figure 1: Timeline of the model

4. Equilibrium without the right to silence

As the benchmark case, this section examines the case without the right to silence. In each period, the expert needs to make a recommendation. Formally, $R_1 = R_2 = R = \{x, y\}$. Let $s_t \in \{x, y\}$ be the signal in period t.

We focus on the pure-strategy perfect Bayesian–Nash equilibria (PBE), in which the expert recommends the signal that she receives at the first opportunity.[14][15] Note that because of our symmetric setup, there is no pandering concern.[16]

Under the restriction that the expert conveys her received signal truthfully during the first opportunity, there are two τ-independent strategies: truthful recommendation and consistent recommendation. Strategy $r = (r_1, r_2)$ is the *truthful recommendation* if the expert recommends the signal that she receives in each period. Precisely, for each $\tau \in T$ and each history $(r_1, s_1, s_2) \in R \times S^2$, $r_1(s_1 \colon \tau) = s_1$ and $r_2(r_1, s_1, s_2 \colon \tau) = s_2$. The other strategy is the *consistent recommendation*. Under this strategy, although the expert truthfully reports her signal in the first period, in the second period, she follows the first-period choice irrespective of the second signal. Precisely, for each $\tau \in T$,

[14] As another possible strategy, the expert recommends the opposite signal from that which she receives at the first opportunity. However, if it is a PBE, the evaluator learns that the reported signal is opposite to the received signal. This is the same as the focused strategy.

[15] Because the first-stage message is cheap talk, a babbling message is also possible, that is, reporting a message that is independent of the received signal. The difference between this and the case where remaining silent is possible is discussed in Section 5.

[16] A pandering concern implies sticking with the ex ante more likely recommendation, even if the expert's signal indicates otherwise.

$r_1(s_1 : \tau) = s_1$ and $r_2(r_1, s_1, s_2 : \tau) = r_1$ if $r_1(s_1 : \tau) = s_1$.[17] On an out-of-the-equilibrium path, that is, if $r_2 \neq r_1$, we assume that the evaluator believes that the expert recommended her signals truthfully at each opportunity.[18] The following proposition summarizes the equilibrium.

Proposition 1. *Suppose that Φ is weakly convex. Then:*

(i) For each $K \geqslant 0$, there exists a $\bar{\alpha}^T > 0$ such that for each $\alpha < \bar{\alpha}^T$, the truthful recommendation is not a PBE.

(ii) For each $K \geqslant 0$, there exists a $\bar{\alpha}^C > 0$ such that for each $\alpha < \bar{\alpha}^C$, the consistent recommendation is a PBE.

(iii) Suppose that neither the truthful nor the consistent recommendation is a PBE. Then, there is a mixed strategy equilibrium in which the expert truthfully reveals her signal in the first stage, and mixes the messages in the second if she receives inconsistent signals.

Proof. See Appendix A.1. □

Under this proposition, with a sufficiently small α, among the equilibria where the expert makes a truthful recommendation at the first opportunity, the expert behaves consistently.

To obtain the intuition behind this proposition, let $\theta_{r_1 r_2 \omega}$ be the expected value of the expert's ability θ_1 with respect to the evaluator's belief after observing recommendations r_1, r_2, and state realization ω.

Suppose that the expert receives signal $s_1 = x$ at the first opportunity and recommends it. In addition, suppose that the expert receives $s_2 = y$ at the second opportunity. Then, under the truthful recommendation, the reputation gain is $\Phi(\theta_{xy\omega})$,

[17] This strategy requires consistency only on the equilibrium path.

[18] Although there are two other off-path beliefs in pure strategies, the consistent recommendation is not a PBE with such beliefs. If the evaluator believes that $s_2 = r_1 \neq r_2$, recommending $r_2 \neq r_1$ involves a reputational gain. Thus, when $s_1 \neq s_2$, the expert deviates to recommend s_2 to obtain monetary gain K. If the evaluator believes that r_2 is independent of s_2, the assessment is also independent of r_2. Thus, when $s_1 \neq s_2$, the expert deviates to recommend s_2 to obtain K.

whereas that under the consistent recommendation is $\Phi(\theta_{xx\omega})$. The key for proving Proposition 1 is that inequality $\Phi(\theta_{xxx}) - \Phi(\theta_{xyx}) > \Phi(\theta_{xyy}) - \Phi(\theta_{xxy})$ holds. The left-hand side is the reputation gain obtained by behaving consistently when the recommendation matches the true state, and the right-hand side is the loss from behaving consistently when the recommendation does not match compared with that from behaving inconsistently. This inequality implies that behaving consistently leads to a higher reputation gain.

To simplify the discussion, consider the case where Φ is linear, that is, $\Phi(\theta) = \theta$. To gain an intuition of inequality $\theta_{xxx} - \theta_{xyx} > \theta_{xyy} - \theta_{xxy}$, consider the case where $\alpha = 0$. Note that $\theta_{xxx} > \theta_{xyy} = \theta_{xyx} > \theta_{xxy}$.[19] This inequality implies that the assessment is highest when both of the received signals match the realized state and lowest when both signals do not match this state. The assessment is intermediate when one signal matches the state but the other does not. Ignoring the prior distribution of the ability, when $s_1 = s_2 = \omega$, $\theta_1 = 1$ maximizes the likelihood of this event. When $s_1 \neq s_2$, $\theta_1 = 1/2$ maximizes the likelihood. Note that because $\theta_1 \geqslant 1/2$, when $s_1 = s_2 \neq \omega$, $\theta_1 = 1/2$ also maximizes the likelihood. Based on these facts, whereas θ_{xxx} becomes close to 1, the other assessments become close to 1/2. Thus, the difference $\theta_{xxx} - \theta_{xyx}$ is larger than $\theta_{xyy} - \theta_{xxy}$.

Example 1. Suppose that $D = \{1/2, 3/4, 1\}$, $f(1/2) = a$, $f(1) = b$, and $f(3/4) = 1 - a - b$. In Figure 2 ($K = 1.2$, $a = b = 0.4$), three curves are drawn. The threshold $\bar{\alpha}^T$ in Proposition 1 is obtained from the intersection of $W(\alpha)$ and $1/(1+\alpha)$, and the threshold $\bar{\alpha}^C$ is obtained from the intersection of $V(\alpha)$ and $1/(1+\alpha)$.

As depicted in Figure 2, $\bar{\alpha}^T$ and $\bar{\alpha}^C$ are around 0.2. Then, if $\alpha < 0.2$, only the consistent recommendation is a PBE.

Remark 1. Notice that, as depicted in Figure 2, $\bar{\alpha}^C < \bar{\alpha}^T$. Thus, if $\alpha \in (\bar{\alpha}^C, \bar{\alpha}^T)$, neither of

[19] The equality $\theta_{xyy} = \theta_{xyx}$ holds even when $\alpha > 0$. See the proof of Proposition 1.

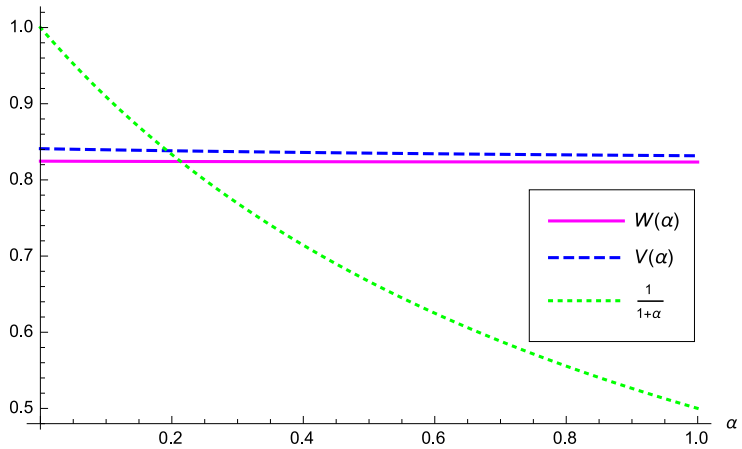

Figure 2: $V(\alpha)$, $W(\alpha)$, and $1/(1+\alpha)$, $a = b = 0.4$, $K = 1.2$.

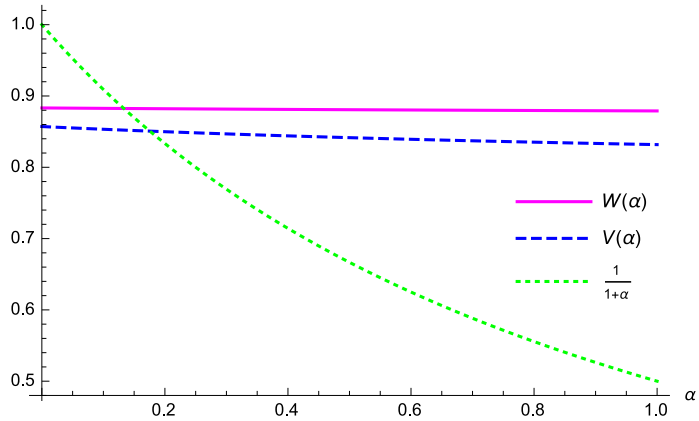

Figure 3: $V(\alpha)$, $W(\alpha)$ and $1/(1+\alpha)$, $a = b = 0.2$, $K = 0.5$.

the consistent and truthful recommendations is a PBE. In this case, as Proposition 1(iii) states, the expert mixes the consistent and truthful recommendations.

In another example, there is a region of α in which both the consistent and truthful recommendations are PBEs (i.e., $\bar{\alpha}^T < \bar{\alpha}^C$. See Figure 3).

5. The right to silence

The previous section focuses on the equilibrium where the expert necessarily recommends some alternative in all periods. However, because accuracy increases, only the second-period signal is necessary for an efficient decision. Thus, there seems to be

no need for the first-period recommendation. Our question is, when the expert has the option to reveal or conceal the information about the signal received in the first period, which option does she choose? To formalize this question, we add a choice to remain silent in the first period. That is, $r_1 \in R_1 = \{x, y, \varnothing\}$, where $r_1 = \varnothing$ implies that the expert remains silent at the first opportunity. Observing this message $r_1 = \varnothing$, we assume that the evaluator places an initial probability of 1/2 on the event that $s_1 = x$. Thus, by sending this message, the expert can conceal her received signal. At the second opportunity, we assume that the expert must recommend one of the two alternatives, that is $r_2 \in R_2 = \{x, y\}$.

5.1. Unknown ability

This subsection focuses on the case where the expert receives no information about her ability θ_1 before the first stage, that is $|T| = 1$. Notice that because of the symmetry of the distribution of the state of the world, the expert learns nothing about her ability from the first signal. Then, each action of the expert in the first stage yields no information about her ability.[20]

To answer the question regarding whether the expert prefers revealing the information about s_1 to concealing it at the first stage, we can consider two strategies. One strategy is revealing the information about s_1. As we saw in the previous section, if the expert recommends her first signal at the first stage, when α is sufficiently small, the consistent recommendation is the unique equilibrium of the continuation game. The other strategy is concealing the information about s_1. If the information about s_1 is concealed at period 1 (i.e., $r_1 = \varnothing$), it is optimal to send s_2 (i.e., $r_2 = s_2$) at period 2 because the second signal is more precise. This strategy is referred to as the *waiting strategy*. The consistent recommendation involves a *snap decision* in the sense that the

[20]This is justified by the "no-signaling-what-you-don't-know" condition (Fudenberg and Tirole, 1991).

expert's decision is based only on the immediate, less accurate signal. In light of this observation, the waiting strategy seems to be a better strategy. However, the following proposition shows that the consistent recommendation has a reputational advantage.

Proposition 2. *Suppose that $K = 0$ and $R_1 = \{x, y, \varnothing\}$. In addition, suppose that the expert receives no information about her ability. Then:*

(i) Assume that Φ is linear. Then, the expected utility gained under the consistent recommendation is the same as that gained under the waiting strategy.

(ii) Assume that Φ is strictly convex and $\Phi'''(a) \geqslant 0$ for each $a \in [1/2, 1]$. In addition, assume that:

$$(m_2^2 - m_3\mu)(1 - \mu) + \mu(\mu - m_2)(m_2 - \mu^2) \leqslant 0, \tag{2}$$

where μ is the mean of θ, m_2 is the second-order moment, and m_3 is the third-order moment about 0 (i.e., $\mu = \int \theta f(\theta) d\theta$, $m_2 = \int \theta^2 f(\theta) d\theta$, and $m_3 = \int \theta^3 f(\theta) d\theta$).

Then, the expected utility under the consistent recommendation strategy is greater than that under the waiting strategy for a sufficiently small $\alpha > 0$. Thus, the consistent recommendation is a PBE.

Proof. See Appendix A.3. □

The intuition is as follows. If the expert remains silent in the first period, her ability is assessed based only on the second signal, whereas when the expert speaks in the first period and employs the consistent recommendation strategy, her ability is assessed based only on the first signal. Under the assumption that $\theta_2 = \frac{(1+\alpha)\theta_1}{1+\alpha\theta_1}$, if the expert waits for a more accurate signal and her recommendation matches the true state, this is partly attributed to the improvement of the signal and less to the expert's ability than in the case with the first signal. By contrast, if the recommendation does not

match the true state, this is attributed to the expert's ability more than in the case with the first signal. In this sense, a more accurate signal about the state is not necessarily more informative about the expert's ability. Therefore, if the expert remains silent, her ability is discounted regardless of whether her recommendation matches the true state.[21] This indicates the merit of the consistent recommendation.

By contrast, under the waiting strategy, the probability of matching the recommendation to the state is greater than that under the consistent recommendation. Because the expert's reputation is improved when her recommendation matches the true state, this is a demerit of the consistent recommendation.

In fact, with a linear Φ, the merit and demerit cancel out. This is because the expert has the same information as the evaluator about her ability at the first stage and, thus, the ex ante expectation of the evaluator's assessment—which is an expectation about the expert's ability—is an iterated expectation of the expert's ability. Therefore, by the law of iterated expectations, under any strategy, the expectation of the assessment is the unconditional expectation $\mu = \int \theta f(\theta) d\theta$.[22]

Thus, which strategy has a reputational advantage depends on the shape of Φ, the utility function for the reputation. Because we assume that Φ is convex, the expert is risk-loving. Proposition 2(ii) provides a sufficient condition for the consistent recommendation to be better. Intuition is the following. This finding reflects the following considerations. The first condition in Proposition 2(ii) indicates that the convexity of Φ increases as the expert' s reputation increases, so the expert effectively becomes more risk-loving as her reputation increases. Regardless of whether the recommendation matches the state, the expert' s reputation is higher when the recommendation is consistent. Therefore, even when the two strategies entail the same risk, the consistent recommendation is preferred. The second condition, inequality (2), guarantees that

[21] Indeed, we can show that $\theta_{xxx} > \theta_{\varnothing xx}$ and $\theta_{xxy} > \theta_{\varnothing xy}$ without any assumption on the distribution function F. See the proof of Proposition 2.

[22] Holmström (1999) finds an equivalent fact using his model.

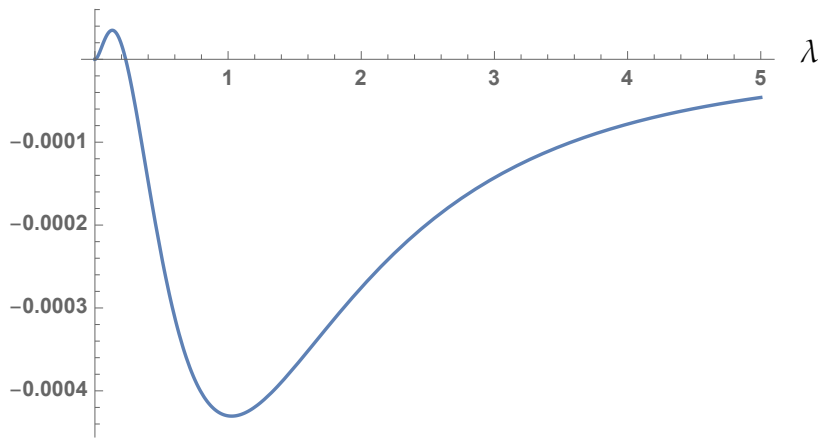

Figure 4: Values of the left-hand side of (2) when $\ln(\theta/(1-\theta))$ is distributed by the exponential distribution with rate λ.

when α is sufficiently small, the reputation gap between success and failure in terms of matching the state under the waiting strategy $\theta_{\varnothing xx} - \theta_{\varnothing xy}$ is smaller than that under the consistent recommendation, $\theta_{xxx} - \theta_{xxy}$ (See the proof of the proposition.) Consequently, the consistent recommendation entails more risk than the waiting strategy. The first and the second conditions together ensure the expert prefers consistency.

Inequality (2) is strictly satisfied when f is a uniform distribution on $[1/2, 1]$. As another example, suppose that $\ln(\theta/(1-\theta))$ is distributed by the exponential distribution with rate λ. Although inequality (2) is violated with a small λ, it is satisfied with a sufficiently large λ (see Figure 4). Notice that the mean of the exponential distribution with rate λ is $1/\lambda$. Then, a lower expected ability implies a larger λ, which means that the expert is more likely to make a snap decision. Intuitively, when the expected ability is lower, because Φ is convex, a small reputation gain makes little difference to utility. Then, the expert prefers the strategy that can yield a larger reputation gain, which is snap decision-making.

Babbling equilibrium and equilibrium selection Note that the first-stage communication is cheap talk and then a babbling equilibrium exists. For example, consider

a strategy such that $r_1(s_1) = x$ for each $s_1 \in \{x, y\}$. The evaluator has a belief that places a probability of $\Pr(s_1 = x \mid r_1) = 1/2$ on each $r_1 \in \{x, y, \varnothing\}$. Then, any recommendation r_1 reveals no information about the received signal. After sending this message, $r_2(s_1, r_1, s_2) = s_2$ is optimal in the second period. Then, this strategy is also a PBE. Nevertheless, Proposition 2(ii) demonstrates the case in which snap decision-making has a reputational advantage. This implies that the equilibrium involving snap decision-making is *neologism-proof* (Farrell, 1993), whereas the babbling equilibrium is not. To see this, first, let us explain the concept. A set of *neologisms* is denoted by $\mathcal{N} = \{N \colon N \subseteq S_1 = \{x, y\}\}$. Let $\mathcal{P}(N)$ be the set of first signals under which the expert wants to convey neologism N, which is defined by:

$$\mathcal{P}(N) = \left\{ s_1 \in S_1 \colon \max_{r_2} E[I(r_2 = \omega)K + \Phi(\mathbb{E}[\theta_1 \mid s_1 \in N, r_2, \omega])] > u^*(s_1) \right\}, \tag{3}$$

where $u^*(s_1)$ is the expected payoff under the focused equilibrium when the first signal is s_1. A neologism is credible if $N = \mathcal{P}(N)$. A PBE is *neologism-proof* if no neologism is credible relative to it.

Relative to the babbling equilibrium, under the assumptions of Proposition 2(ii), $\{x\}$ and $\{y\}$ are credible neologisms; with those neologisms, the left-hand side of the inequality inside the braces in (3) is the expected utility under the consistent recommendation. Because the expected utility under the babbling equilibrium is the same as that under the waiting strategy, the babbling equilibrium is not neologism-proof. By contrast, for the same reason, relative to the consistent recommendation, no neologism is credible. Thus, the equilibrium involving snap decision-making survives the neologism-proof criterion. In this sense, snap decision-making is a unique credible behavior under the neologism-proof criterion.

5.2. Known ability

This subsection considers the case where the expert knows her ability, that is, $\tau = \theta_1$, $f(\theta \mid \theta) = 1$, and $f(\theta' \mid \theta) = 0$ for each $\theta \neq \theta'$. As in the previous section, we compare two strategies: the waiting strategy and the consistent recommendation strategy.

First, by selecting the out-of-equilibrium belief accordingly, both pooling to wait and pooling to make a snap decision are equilibria.

Proposition 3. *Assume that* $\inf D \in D$.

(i) The waiting strategy is a PBE. The out-of-equilibrium belief places probability 1 *on* $\theta_1 = \inf D$.

(ii) For a sufficiently small α*, the consistent recommendation is a PBE. The out-of-equilibrium belief places probability* 1 *on* $\theta_1 = \inf D$.

Proof. See Appendix A.4 □

In each equilibrium, the out-of-equilibrium belief is natural, as we show using the following lemma:

Lemma 1. *Let* $\mathbb{E}U(\theta\colon r_1 = \varnothing, \beta)$ *be the expected utility of the waiting strategy, and let* $\mathbb{E}U(\theta : r_1 = s_1, \beta)$ *be that of the consistent recommendation strategy when* $\theta_1 = \theta$.

(i) Consider the PBE given in Proposition 3(i). Then, if α *is sufficiently small,* $\mathbb{E}U(\theta\colon r_1 = \varnothing, \beta) - \mathbb{E}U(\theta\colon r_1 = s_1, \beta)$ *is increasing in* θ.

(ii) Consider the PBE given in Proposition 3(ii). Then, $\mathbb{E}U(\theta\colon r_1 = \varnothing, \beta) - \mathbb{E}U(\theta\colon r_1 = s_1, \beta)$ *is decreasing in* θ.

Proof. See Appendix A.4 □

Consider case (i). In this PBE, the expert with $\theta_1 = \inf D$ is most likely to deviate to $r_1 \neq \varnothing$. Consider case (ii). If $r_1 = \varnothing$, the evaluator believes that $\theta_1 = \inf D$. Under this belief, Lemma 1(ii) shows that the expert with $\theta_1 = \inf D$ is most likely to deviate

to $r_1 = \varnothing$. In this sense, the out-of-equilibrium beliefs in Proposition 3 are consistent with the spirit of the intuitive criterion (Cho and Kreps, 1987).

Finally, in this section, let us discuss the existence of a separating equilibrium associated with the expert's ability. In fact, with a sufficiently small α, there is no pure-strategy separating equilibrium.

Proposition 4. *For a sufficiently small $\alpha > 0$, no equilibrium satisfies the following conditions: (1) there is a nonempty set $D_\varnothing \subsetneq D$, (2) the expert with $\theta \in D_\varnothing$ employs the waiting strategy, and (3) the expert with $\theta \in D \setminus D_\varnothing$ employs the consistent recommendation strategy.*

Proof. See Appendix A.5. □

The intuition is as follows. Compared with the waiting strategy, the advantage of the consistent recommendation strategy lies only in the reputation term. Thus, the reputation gain for the consistent recommendation must be higher than that for the waiting strategy. For the existence of a separating equilibrium, the waiting strategy must have an advantage in terms of the probability of a correct decision, the benefit of which is increasing in K. However, for a sufficiently large K to separate the behaviors, the consistent behavior is not optimal at the second stage. For this reason, no equilibrium separates the waiting and consistent recommendation strategies.

5.3. Informativeness of the expert's strategies

Finally, we compare the informativeness of the expert's strategies about the state of the world in the sense of Blackwell. The most-informative strategy is the truthful recommendation because it reveals all the signals that the expert receives. The second-most-informative strategy is the waiting strategy, and the third is the consistent recommendation strategy. This is because the second signal is more informative than the first signal. Because these strategies reveal only one signal, they are less informative than the truthful recommendation. The least informative strategy is sending

babbling messages at each opportunity, that is, recommending one alternative independent of the received signals, s_1 and s_2. Proposition 2 shows that when we focus on neologism-proof equilibria, under the given conditions, the snap decision, the second-least-informative strategy, is the unique prediction.

6. Extension

This section briefly presents a few extensions of our model for consistent behavior. The formal analyses are given in Appendix B. First, in the basic model, the growth ratio is common to all θ and the value is known. We can extend our analysis to a more general setting. Suppose that the growth ratio α is a function of θ—say, $\alpha(\theta)$. We show that if $\alpha(\theta)$ satisfies some reasonable conditions, and if $\max_{\theta \in D} \alpha(\theta)$ is sufficiently small, then the consistent recommendation is a PBE. For example, if θ_2 is written as $\frac{\theta_2}{1-\theta_2} = \left(\frac{\theta_1}{1-\theta_1}\right)^{1+\alpha}$, our derived condition is satisfied when α is sufficiently small. The consistent behavior remains robust when α is independent of θ but is randomly drawn. When α is randomly drawn, it is reasonable to consider the case in which the expert is concerned with the reputation of her α. In this case, however, unlike the case involving the expert's reputation for θ, the truthful recommendation is always a PBE.

For another type of extension, one may wonder whether this kind of consistent behavior can be observed if two or more experts competitively and simultaneously recommend a choice to a client. In fact, there are negative and positive effects of consistency. One is statistical herding (Banerjee, 1992 and Bikhchandani, Hirshleifer and Welch, 1992): when the two experts' opinions differ from each other, each expert loses confidence about her first opinion and thus, each may change her opinion. However, this incentive is mitigated by the *anti*-herding incentive (Avery and Chevalier, 1999, Effinger and Polborn, 2001, Levy, 2004). That is, if an expert changes her first opinion

and follows that of the counterpart, this would reveal that the expert's ability is low. This signaling incentive strengthens the incentive for consistent behavior. Therefore, the competition effect on consistency becomes ambiguous.

7. Conclusion

This study investigates the incentive to behave consistently. Although models of consistent behavior are proposed by authors such as Li (2007), Sabourian and Sibert (2009), and FZ, this study not only extends their results but also shows a novel result, which we refer to as snap decision-making. The combination of consistent behavior and snap decision-making makes the expert's recommendation less valuable. If the expert is a politician, a poor decision could result in significant costs to society. As a remedy, our study suggests that if the expert can obtain a more accurate signal in the future, the client should silence the expert at the first opportunity by imposing a cost for breaking this silence too early. Creating a norm that encourages deliberation is an example. A normative conscience or peer pressure imposes a cost on making a snap decision. In the political context, the media should criticize a politician for declaring her intentions too early, although care should be taken that noncommittal statements are not criticized.

This study has certain limitations that point to future research possibilities. One is that the signals and actions that we discussed are binary. Future work should aim to extend our model to a continuous one and investigate the incentives involved in making snap decisions. In addition, we employ a specific functional form to capture the improvement in signal accuracy. Although we consider an extension in Section 6, a complete characterization of the conditions under which consistent recommendations arise in equilibrium remains to be determined. Furthermore, Proposition 2(ii) only provides a sufficient condition for snap decision-making to be optimal. Consequently,

the full set of conditions under which the waiting strategy is optimal remains to be specified. Another limitation is that reputation concern in our model only references the expectation of ability. Ottaviani and Sørensen (2006a) consider a more general form of reputation concern that would be interesting to apply to our model.

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A. Omitted proofs

A.1. Proof of Proposition 1

Part 1. We show that the truthful recommendation is not a PBE.

Consider the second opportunity. Let $p_{s_1 s_2}$ be the posterior belief of the expert regarding the event that $\omega = x$ when she receives signals s_1 and s_2, which is calculated as:

$$
\begin{aligned}
p_{xx} &= \int \frac{\theta_1\theta_2}{\theta_1\theta_2 + (1-\theta_1)(1-\theta_2)} f(\theta_1 \mid \tau, s_1 = s_2 = x) d\theta_1, \\
p_{xy} &= \int \frac{\theta_1(1-\theta_2)}{\theta_1(1-\theta_2) + \theta_2(1-\theta_1)} f(\theta_1 \mid \tau, s_1 = x, s_2 = y) d\theta_1, \\
p_{yx} &= \int \frac{(1-\theta_1)\theta_2}{\theta_1(1-\theta_2) + \theta_2(1-\theta_1)} f(\theta_1 \mid \tau, s_1 = y, s_2 = x) d\theta_1, \\
p_{yy} &= \int \frac{(1-\theta_1)(1-\theta_2)}{\theta_1\theta_2 + (1-\theta_1)(1-\theta_2)} f(\theta_1 \mid \tau, s_1 = s_2 = y) d\theta_1.
\end{aligned}
$$

Note that $p_{xx} > p_{yx} > 1/2 > p_{xy} > p_{yy}$.

Suppose that $r_1 = s_1 = x$ and $s_2 = y$. Then, the truthful recommendation (i.e., recommending y) is optimal when:

$$p_{xy}\Phi(\theta^T_{xyx}) + (1 - p_{xy})(\Phi(\theta^T_{xyy}) + K) \geqslant p_{xy}(\Phi(\theta^T_{xxx}) + K) + (1 - p_{xy})\Phi(\theta^T_{xxy}), \quad \text{(A.1)}$$

where $\theta^T_{r_1 r_2 \omega}$ is the ex post expected value of the expert's ability θ_1 with respect to the evaluator's belief after observing recommendations r_1, r_2 and state realization ω. Superscript T indicates the assessment under the truthful recommendation. The values of $\theta^T_{r_1 r_2 \omega}$ are given by:

$$\begin{aligned}
\theta^T_{xxx} = \theta^T_{yyy} &= \frac{\int \theta^2 \theta_2 f(\theta) d\theta}{\int \theta \theta_2 f(\theta) d\theta} = \frac{\int \frac{\theta^3}{1+\alpha\theta} f(\theta) d\theta}{\int \frac{\theta^2}{1+\alpha\theta} f(\theta) d\theta} \\
\theta^T_{xxy} = \theta^T_{yyx} &= \frac{\int \theta \frac{(1-\theta)^2}{1+\alpha\theta} f(\theta) d\theta}{\int \frac{(1-\theta)^2}{1+\alpha\theta} f(\theta) d\theta} \\
\theta^T_{xyx} = \theta^T_{xyy} = \theta^T_{yxx} &= \theta^T_{yxy} = \frac{\int \frac{\theta^2(1-\theta)}{1+\alpha\theta} f(\theta) d\theta}{\int \frac{\theta(1-\theta)}{1+\alpha\theta} f(\theta) d\theta}.
\end{aligned} \quad \text{(A.2)}$$

The left-hand side of (A.1) is the expected utility of recommending y and the right-hand side is that of recommending x. Therefore, recommending y is optimal if and only if:

$$\frac{p_{xy}}{1 - p_{xy}} = \frac{1}{1 + \alpha} \leqslant \frac{\Phi(\theta^T_{xyy}) - \Phi(\theta^T_{xxy}) + K}{\Phi(\theta^T_{xxx}) - \Phi(\theta^T_{xyx}) + K}. \quad \text{(A.3)}$$

To verify whether the above inequality holds, we provide the following lemma.

Lemma 2. *For each $\alpha \geqslant 0$, $\theta^T_{xxx} - \theta^T_{xyx} > \theta^T_{xyy} - \theta^T_{xxy}$. Thus, if Φ is weakly convex, $\Phi(\theta^T_{xxx}) - \Phi(\theta^T_{xyx}) > \Phi(\theta^T_{xyy}) - \Phi(\theta^T_{xxy})$.*

Proof. See Appendix A.2. □

By this inequality, with a sufficiently small α, inequality (A.3) is violated. □

Part 2. We show that the consistent recommendation is a PBE.

Consider the second opportunity. Suppose that $r_1 = s_1 = x$. As in Part 1, at the second opportunity, recommending x is optimal when:

$$\frac{p_{xx}}{1-p_{xx}} \geqslant \frac{\Phi(\theta^C_{xyy}) - \Phi(\theta^C_{xxy}) + K}{\Phi(\theta^C_{xxx}) - \Phi(\theta^C_{xyx}) + K} \quad \text{if } s_2 = x, \tag{A.4}$$

$$\frac{p_{xy}}{1-p_{xy}} \geqslant \frac{\Phi(\theta^C_{xyy}) - \Phi(\theta^C_{xxy}) + K}{\Phi(\theta^C_{xxx}) - \Phi(\theta^C_{xyx}) + K} \quad \text{if } s_2 = y, \tag{A.5}$$

where $\theta^C_{r_1 r_2 \omega}$ is the ex post expected value of the expert's ability θ_1 under a consistent recommendation. Superscript C indicates that it is the assessment under the consistent recommendation. The value of $\theta^C_{r_1 r_2 \omega}$ is given as follows:

$$\begin{aligned}
\theta^C_{xxx} &= \theta^C_{yyy} = \frac{\int \theta^2 f(\theta) d\theta}{\int \theta f(\theta) d\theta}, \quad \theta^C_{xxy} = \theta^C_{yyx} = \frac{\int \theta(1-\theta) f(\theta) d\theta}{\int (1-\theta) f(\theta) d\theta}, \\
\theta^C_{xyx} &= \theta^C_{yxy} = \frac{\int \theta^2 (1-\theta_2) f(\theta) d\theta}{\int \theta(1-\theta_2) f(\theta) d\theta} = \frac{\int \frac{\theta^2(1-\theta)}{1+\alpha\theta} f(\theta) d\theta}{\int \frac{\theta(1-\theta)}{1+\alpha\theta} f(\theta) d\theta}, \\
\theta^C_{xyy} &= \theta^C_{yxx} = \frac{\int \theta\theta_2 (1-\theta) f(\theta) d\theta}{\int \theta_2 (1-\theta) f(\theta) d\theta} = \frac{\int \frac{\theta^2(1-\theta)}{1+\alpha\theta} f(\theta) d\theta}{\int \frac{\theta(1-\theta)}{1+\alpha\theta} f(\theta) d\theta} = \theta^C_{xyx},
\end{aligned} \tag{A.6}$$

The following lemma is helpful to verify inequalities (A.4) and (A.5).

Lemma 3. *Under the consistent recommendation, $\theta^C_{xxx} - \theta^C_{xyx} > \theta^C_{xyy} - \theta^C_{xxy}$ for each $\alpha \geqslant 0$. Thus, if Φ is weakly convex, $\Phi(\theta^C_{xxx}) - \Phi(\theta^C_{xyx}) > \Phi(\theta^C_{xyy}) - \Phi(\theta^C_{xxy})$.*

Proof. See Appendix A.2. □

Recall that:

$$\begin{aligned}
p_{xx} &= \int \frac{\theta\theta_2}{\theta\theta_2 + (1-\theta)(1-\theta_2)} f(\theta \mid \tau, s_1 = s_2 = x) d\theta \geqslant 1/2, \\
p_{xy} &= \int \frac{\theta(1-\theta_2)}{\theta(1-\theta_2) + \theta_2(1-\theta)} f(\theta \mid \tau, s_1 = x, s_2 = y) d\theta = \frac{1}{2+\alpha}.
\end{aligned}$$

Because $p_{xx}/(1-p_{xx}) \geqslant 1$ and $p_{xy}/(1-p_{xy}) = 1/(1+\alpha)$, recommending x is optimal

if and only if (A.5) holds. From Lemma 3, with a sufficiently small α, (A.5) holds.

Consider the first stage. Without loss of generality, we assume that $s_1 = x$. To show that recommending x is optimal at the first opportunity, let $r_2(y, x, s_2)$ be the continuation strategy when $r_1 = y$. Then, we have the following four cases:

Case 1. $r_2(y, x, s_2) = x$ *for each* $s_2 \in \{x, y\}$. Let $S(\theta) = \theta\theta_2 + (1-\theta)(1-\theta_2)$, which is the probability that the expert draws the same signals. Then, recommending x at the first stage is optimal if and only if:

$$\begin{aligned}
&\int \big[S(\theta)(p^{\theta}_{xx}(\Phi(\theta^C_{xxx}) + K) + (1 - p^{\theta}_{xx})\Phi(\theta^C_{xxy})) \\
&+ (1 - S(\theta))(p^{\theta}_{xy}(\Phi(\theta^C_{xxx}) + K) + (1 - p^{\theta}_{xy})\Phi(\theta^C_{xxy}))\big] f(\theta \mid \tau) d\theta \\
&> \int \big[S(\theta)(p^{\theta}_{xx}(\Phi(\theta^C_{yxx}) + K) + (1 - p^{\theta}_{xx})\Phi(\theta^C_{yxy})) \\
&+ (1 - S(\theta))(p^{\theta}_{xy}(\Phi(\theta^C_{yxx}) + K) + (1 - p^{\theta}_{xy})\Phi(\theta^C_{yxy}))\big] f(\theta \mid \tau) d\theta,
\end{aligned}$$

where $p^{\theta}_{xx} = \frac{\theta\theta_2}{\theta\theta_2 + (1-\theta)(1-\theta_2)}$, and $p^{\theta}_{xy} = \frac{\theta(1-\theta_2)}{\theta(1-\theta_2)+\theta(1-\theta_2)}$. Equivalently, from (A.6):

$$\int [\theta(\Phi(\theta^C_{xxx}) - \Phi(\theta^C_{xyy})) + (1-\theta)(\Phi(\theta^C_{xxy}) - \Phi(\theta^C_{xyx}))] f(\theta \mid \tau) d\theta > 0.$$

Because $\theta \geqslant 1/2$ and $\Phi(\theta^C_{xxx}) - \Phi(\theta^C_{xyy}) > \Phi(\theta^C_{xyx}) - \Phi(\theta^C_{xxy})$ from Lemma 3, this inequality holds.

Case 2. $r_2(y, x, s_2) = y$ *for each* $s_2 \in \{x, y\}$. As in Case 1, recommending x is optimal if and only if:

$$(\Phi(\theta^C_{xxx}) - \Phi(\theta^C_{xxy}) + K) \int (2\theta - 1) f(\theta \mid \tau) > 0.$$

Because $\theta^C_{xxx} > \theta^C_{xxy}$ and $\theta > 1/2$, the above inequality holds.

Case 3. $r_2(y,x,x) = x$ and $r_2(y,x,y) = y$. Note that from (A.5), the expected utility of recommending x satisfies:

$$\begin{aligned}
&\int \big[S(\theta)(p^{\theta}_{xx}(\Phi(\theta^C_{xxx}) + K) + (1 - p^{\theta}_{xx})\Phi(\theta^C_{xxy})) \\
&+ (1 - S(\theta))(p^{\theta}_{xy}(\Phi(\theta^C_{xxx}) + K) + (1 - p^{\theta}_{xy})\Phi(\theta^C_{xxy}))\big] f(\theta \mid \tau) d\theta \\
&\geqslant \int \big[S(\theta)(p^{\theta}_{xx}(\Phi(\theta^C_{xxx}) + K) + (1 - p^{\theta}_{xx})\Phi(\theta^C_{xxy})) \\
&+ (1 - S(\theta))(p^{\theta}_{xy}\Phi(\theta^C_{xyx}) + (1 - p^{\theta}_{xy})(\Phi(\theta^C_{xyy}) + K)\big] f(\theta \mid \tau) d\theta
\end{aligned}$$

Then, recommending x is optimal if:

$$\begin{aligned}
&\int \big[S(\theta)(p^{\theta}_{xx}(\Phi(\theta^C_{xxx}) + K) + (1 - p^{\theta}_{xx})\Phi(\theta^C_{xxy})) \\
&+ (1 - S(\theta))(p^{\theta}_{xy}\Phi(\theta^C_{xyx}) + (1 - p^{\theta}_{xy})(\Phi(\theta^C_{xyy}) + K)\big] f(\theta \mid \tau) d\theta \\
&\geqslant \int \big[S(\theta)(p^{\theta}_{xx}(\Phi(\theta_{yxx}) + K) + (1 - p^{\theta}_{xx})\Phi(\theta^C_{yxy})) \\
&+ (1 - S(\theta))(p^{\theta}_{xy}\Phi(\theta^C_{yyx}) + (1 - p^{\theta}_{xy})(\Phi(\theta^C_{yyy}) + K)\big] f(\theta \mid \tau) d\theta.
\end{aligned}$$

Equivalently, from (A.6), we have:

$$\int [(\Phi(\theta^C_{xxx}) - \Phi(\theta^C_{xxy}))\theta(2\theta_2 - 1) + (1 - \theta_2)(\Phi(\theta^C_{xyx}) - \Phi(\theta^C_{xxy}))(2\theta - 1)] f(\theta \mid \tau) d\theta > 0.$$

This inequality holds because $\theta^C_{xxx} > \theta^C_{xyy}$, $\theta^C_{xyx} > \theta^C_{xxy}$, $\theta > 1/2$, and $\theta_2 > 1/2$.

Case 4. $r_2(y,x,x) = y$ and $r_2(y,x,y) = x$. Using the same logic as that for Case 3, recommending x is optimal if:

$$\int \big[(\theta + \theta_2 - 1)(\Phi(\theta^C_{xxx}) - \Phi(\theta^C_{xxy})) + K(2\theta_2 - 1)\big] f(\theta \mid \tau) d\theta \geqslant 0.$$

As in Case 3, this inequality holds.

In any case, recommending x is optimal and, thus, the consistent recommendation is an equilibrium. □

Part 3. Finally, we consider the case in which neither the truthful nor the consistent recommendation is a PBE. Then, we show that a mixed strategy equilibrium exists. Without loss of generality, suppose that $r_1 = s_1 = x$. Now, consider the following continuation strategy in the second stage: if $s_2 = y$, the expert recommends x with probability η and y with the complement probability; otherwise, she recommends x. If $s_2 = y$, recommending x is optimal if and only if:

$$\frac{1}{1+\alpha} \geqslant \frac{\Phi(\theta^{\eta}_{xyy}) - \Phi(\theta^{\eta}_{xxy}) + K}{\Phi(\theta^{\eta}_{xxx}) - \Phi(\theta^{\eta}_{xyx}) + K},$$

where:

$$\theta^{\eta}_{xxx} = \frac{\int \theta^2(\theta_2+(1-\theta_2)\eta)f(\theta)d\theta}{\int \theta(\theta_2+(1-\theta_2)\eta)f(\theta)d\theta}, \quad \theta^{\eta}_{xxy} = \frac{\int \theta(1-\theta)((1-\theta_2)+\theta_2\eta)f(\theta)d\theta}{\int (1-\theta)((1-\theta_2)+\theta_2\eta)f(\theta)d\theta}$$
$$\theta^{\eta}_{xyx} = \frac{\int \theta^2(1-\theta_2)f(\theta)d\theta}{\int \theta(1-\theta_2)f(\theta)d\theta}, \quad \theta^{\eta}_{xyy} = \frac{\int \theta(1-\theta)\theta_2 f(\theta)d\theta}{\int (1-\theta)\theta_2 f(\theta)d\theta}$$

It is easy to show that: $\theta^{0}_{r_1 r_2 \omega} = \theta^{C}_{r_1 r_2 \omega}$ and $\theta^{1}_{r_1 r_2 \omega} = \theta^{T}_{r_1 r_2 \omega}$ for each r_1, r_2, and ω. Recall that, as shown in Parts 1 and 2, because neither the truthful nor the consistent recommendation is a PBE, we have the following inequality:

$$\frac{\Phi(\theta^{C}_{xyy}) - \Phi(\theta^{C}_{xxy}) + K}{\Phi(\theta^{C}_{xxx}) - \Phi(\theta^{C}_{xyx}) + K} > \frac{1}{1+\alpha} > \frac{\Phi(\theta^{T}_{xyy}) - \Phi(\theta^{T}_{xxy}) + K}{\Phi(\theta^{T}_{xxx}) - \Phi(\theta^{T}_{xyx}) + K} \tag{A.7}$$

Each $\theta^{\eta}_{r_1 r_2 \omega}$ is continuous in η and, from (A.7), the intermediate value theorem implies that there is an η^* that satisfies:

$$\frac{1}{1+\alpha} = \frac{\Phi(\theta^{\eta^*}_{xyy}) - \Phi(\theta^{\eta^*}_{xxy}) + K}{\Phi(\theta^{\eta^*}_{xxx}) - \Phi(\theta^{\eta^*}_{xyx}) + K}.$$

With this value, the expert is indifferent between recommending x and y.

On the other hand, if $s_2 = x$, recommending x is optimal when:

$$\frac{p_{xx}}{1-p_{xx}} > \frac{\Phi(\theta_{xyy}^{\eta^*}) - \Phi(\theta_{xxy}^{\eta^*}) + K}{\Phi(\theta_{xxx}^{\eta^*}) - \Phi(\theta_{xyx}^{\eta^*}) + K}.$$

Note that $p_{xx}/(1-p_{xx}) > 1/(1+\alpha)$. Thus, the considering strategy profile is a PBE of the second stage.

As for the first-stage behavior, based on the same discussion as that for the proof of Proposition 1(ii), we can show that truth-telling is optimal. □

A.2. Proof of lemmas in Appendix A.1

Proof of Lemma 2. We define that $\varphi(\theta) = \frac{\frac{1}{1+\alpha\theta} f(\theta)}{\int \frac{1}{1+\alpha\theta'} f(\theta')d\theta'}$, $\tilde{\mu} = \int \theta\varphi(\theta)d\theta$, $\tilde{m}_2 = \int \theta^2\varphi(\theta)d\theta$ and $\tilde{m}_3 = \int \theta^3\varphi(\theta)d\theta$, which are, respectively, the first, second, and third moments of θ with density φ. Then, as long as $|D| \geqslant 2$, for each $\alpha \geqslant 0$, from the mathematical facts, $\tilde{m}_3\tilde{\mu} \geqslant \tilde{m}_2^2$ and $\tilde{m}_2 > \tilde{\mu}^2$. Note that because $\theta \in [1/2, 1]$, $\tilde{m}_2 \geqslant \tilde{m}_3$ and $\tilde{\mu} > 1/2$. Then, we can calculate that:

$$\theta_{xxx}^T - \theta_{xyx}^T - (\theta_{xyy}^T - \theta_{xxy}^T) = \frac{\tilde{m}_3\tilde{\mu} - \tilde{m}_2^2 + (2\tilde{\mu}-1)\tilde{m}_2^2 + \tilde{m}_3(\tilde{m}_2 - \tilde{\mu}^2) + \tilde{\mu}^2(\tilde{m}_2 - \tilde{m}_3)}{(\tilde{\mu} - \tilde{m}_2)(1 - \tilde{\mu} - (\tilde{\mu} - \tilde{m}_2))\tilde{m}_2} > 0. \tag{A.8}$$

□

Proof of Lemma 3. Let $\mu = \int \theta f(\theta)d\theta$, $m_2 = \int \theta^2 f(\theta)d\theta$, and $m_3 = \int \theta^3 f(\theta)d\theta$. Under the consistent recommendation, the difference becomes:

$$\theta_{xxx}^C - \theta_{xyx}^C - (\theta_{xyy}^C - \theta_{xxy}^C) = \frac{\int \frac{\theta(1-\theta)}{1+\alpha\theta}((1-2\mu)m_2 + \mu^2 - 2\mu(1-\mu)\theta)f(\theta)d\theta}{\mu(1-\mu)\int \frac{\theta(1-\theta)}{1+\alpha\theta} f(\theta)d\theta}.$$

To show that the above equation is positive, note that:

$$\int \theta(1-\theta)[(1-2\mu)m_2 + \mu^2 - 2\mu(1-\mu)\theta]f(\theta)d\theta$$
$$= (1-\mu)\mu((2\mu-1)\sigma + 2\xi) + \sigma^2\underbrace{(2\mu-1)}_{\text{positive}} > 0,$$

where $\sigma = \int(\theta-\mu)^2 f(\theta)d\theta = m_2 - \mu^2$ and $\xi = \int(\theta-\mu)^3 f(\theta)d\theta = m_3 - 3\mu m_2 + 2\mu^3 = m_3 - \mu m_2 - 2\mu\sigma$. The last inequality follows from the facts that $\mu > 1/2$ and that:

$$(2\mu-1)\sigma + 2\xi = \int(\theta-\mu)^2\underbrace{(2\theta-1)}_{\text{positive}}f(\theta)d\theta > 0.$$

Let $\tilde{\theta} = \frac{(1-2\mu)m_2+\mu^2}{2\mu(1-\mu)}$. Note that $(1-2\mu)m_2 + \mu^2 - 2\mu(1-\mu)\theta < 0$ if and only if $\theta > \tilde{\theta}$. Because $\int \theta(1-\theta)[(1-2\mu)m_2 + \mu^2 - 2\mu(1-\mu)\theta]f(\theta)d\theta > 0$, $\tilde{\theta} > \inf D$. Then, because $1/(1+\alpha\theta)$ is nonincreasing in θ:

$$\int \frac{\theta(1-\theta)}{1+\alpha\theta}((1-2\mu)m_2 + \mu^2 - 2\mu(1-\mu)\theta)f(\theta)d\theta$$
$$\geqslant \frac{\int(\theta(1-\theta))((1-2\mu)m_2 + \mu^2 - 2\mu(1-\mu)\theta)f(\theta)d\theta}{1+\alpha\tilde{\theta}} > 0,$$

which shows that $\theta^C_{xxx} - \theta^C_{xyx} > \theta^C_{xyy} - \theta^C_{xxy}$. This inequality holds even when $\alpha = 0$. □

A.3. Proof of Proposition 2

The waiting strategy is better than the consistent recommendation if and only if:

$$H(\alpha,\Phi) := \int[-[\theta\Phi(\theta_{xxx}) + (1-\theta)\Phi(\theta_{xxy})] + [\theta_2\Phi(\theta_{\varnothing xx}) + (1-\theta_2)\Phi(\theta_{\varnothing xy})]]f(\theta)d\theta$$
$$= -(\mu\Phi(\theta_{xxx}) + (1-\mu)\Phi(\theta_{xxy})) + \mu^*\Phi(\theta_{\varnothing xx}) + (1-\mu^*)\Phi(\theta_{\varnothing xy}) > 0,$$

where $\mu^* = \int \theta_2 f(\theta) d\theta$, and:

$$\theta_{xxx} = \frac{\int \theta^2 f(\theta) d\theta}{\int \theta f(\theta) d\theta}, \qquad \theta_{xxy} = \frac{\int \theta(1-\theta) f(\theta) d\theta}{\int (1-\theta) f(\theta) d\theta},$$
$$\theta_{\varnothing xx} = \frac{\int \theta \frac{(1+\alpha)\theta}{1+\alpha\theta} f(\theta) d\theta}{\int \frac{(1+\alpha)\theta}{1+\alpha\theta} f(\theta) d\theta}, \quad \theta_{\varnothing xy} = \frac{\int \theta \frac{1-\theta}{1+\alpha\theta} f(\theta) d\theta}{\int \frac{1-\theta}{1+\alpha\theta} f(\theta) d\theta}.$$

By a calculation:

$$\mu\theta_{xxx} + (1-\mu)\theta_{xxy} = \mu^*\theta_{\varnothing xx} + (1-\mu^*)\theta_{\varnothing xy}. \tag{A.9}$$

When Φ is linear, we can show that $H(\alpha, \Phi) = 0$ for each α. This proves the first part of the proposition.

Now, we consider the case in which Φ is strictly convex, and prove that $H(\alpha, \Phi) < 0$ for a sufficiently small $\alpha > 0$. Note that:

$$\frac{\partial \theta_{\varnothing xx}}{\partial \alpha} = -\frac{\int \frac{\theta^3}{(1+\alpha\theta)^2} f(\theta) d\theta \int \frac{\theta}{1+\alpha\theta} f(\theta) d\theta - \int \frac{\theta^2}{1+\alpha\theta} f(\theta) d\theta \int \frac{\theta^2}{(1+\alpha\theta)^2} f(\theta) d\theta}{(\int \frac{\theta}{1+\alpha\theta} f(\theta) d\theta)^2} < 0,$$
$$\frac{\partial \theta_{\varnothing xy}}{\partial \alpha} = \frac{\int \frac{\theta(1-\theta)}{(1+\alpha\theta)^2} f(\theta) d\theta \int \frac{\theta(1-\theta)}{1+\alpha\theta} f(\theta) d\theta - \int \frac{\theta^2(1-\theta)}{(1+\alpha\theta)^2} \int \frac{1-\theta}{1+\alpha\theta} f(\theta) d\theta}{(\int \frac{1-\theta}{1+\alpha\theta} f(\theta) d\theta)^2} < 0.$$

We can show that when $\alpha = 0$, because it implies that $\theta_2 = \theta$, $\theta_{\varnothing xx} = \theta_{xxx}$ and $\theta_{\varnothing xy} = \theta_{xxy}$. Then, we have that $\theta_{xxx} > \theta_{\varnothing xx}$ and $\theta_{xxy} > \theta_{\varnothing xy}$ for each $\alpha > 0$. Further, because $\theta_{xxx} > \theta_{xxy}$, we have that $\theta_{\varnothing xx} > \theta_{xxy}$ for a sufficiently small α.

Moreover, by the assumption that inequality (2) holds:

$$\left[\frac{\partial \theta_{\varnothing xx}}{\partial \alpha} - \frac{\partial \theta_{\varnothing xy}}{\partial \alpha}\right]\Bigg|_{\alpha=0} = \frac{(m_2^2 - m_3\mu)(1-\mu) + \mu(\mu - m_2)(m_2 - \mu^2)}{\mu^2(1-\mu)^2} \leqslant 0, \tag{A.10}$$

which implies that $\theta_{xxx} - \theta_{xxy} \geqslant \theta_{\varnothing xx} - \theta_{\varnothing xy}$ for a sufficiently small $\alpha > 0$.

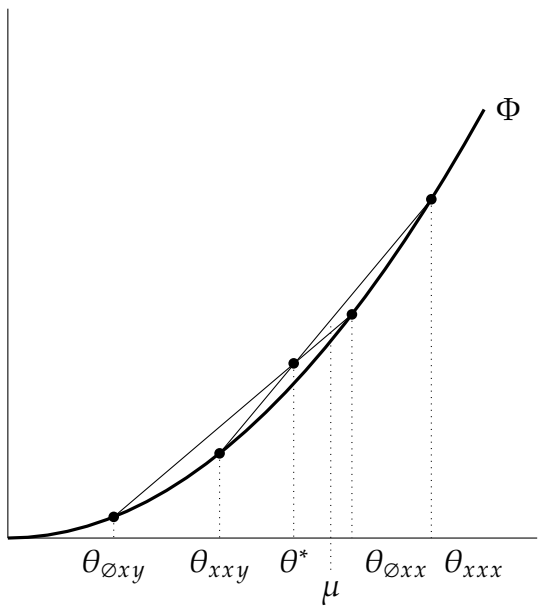

Figure 5: Illustration of θ^*

For each $a \in (0,1)$, let $\gamma(a)$ satisfy:

$$\gamma(a)\theta_{\varnothing xx} + (1-\gamma(a))\theta_{\varnothing xy} = a\theta_{xxx} + (1-a)\theta_{xxy}.$$

Take a small $\alpha > 0$ so that $\theta_{xxx} - \theta_{xxy} \geqslant \theta_{\varnothing xx} - \theta_{\varnothing xy}$. Then, from the inequality $\theta_{xxy} > \theta_{\varnothing xy}$, $\gamma(a) > a$ for each $\alpha \in (0,1)$.

Let Γ be the function satisfying:

$$\Gamma(a) := \gamma(a)\Phi(\theta_{\varnothing xx}) + (1-\gamma(a))\Phi(\theta_{\varnothing xy}) - \left(a\Phi(\theta_{xxx}) + (1-a)\Phi(\theta_{xxy})\right).$$

We can find an a^* satisfying $\Gamma(a^*) = 0$. Let $\theta^* = a^*\theta_{xxx} + (1-a^*)\theta_{xxy} = \gamma(a^*)\theta_{\varnothing xx} + (1-\gamma(a^*))\theta_{\varnothing xy}$. See Figure 5 for an illustration. We now prove that $\mu > \theta^*$.

Claim 1. *For a sufficiently small $\alpha > 0$, $\mu > \theta^*$.*

Proof of Claim 1. Note that: $\theta_{xxx} - \mu = \frac{m_2-\mu^2}{\mu}$, and $\mu - \theta_{xxy} = \frac{m_2-\mu^2}{1-\mu}$. Thus, because

$\mu > 1/2$:

$$\mu > \frac{\theta_{xxx} + \theta_{xxy}}{2}. \tag{A.11}$$

Then, to complete the proof, it is sufficient to show that $\frac{\theta_{xxx}+\theta_{xxy}}{2} \geqslant \theta^*$.

A calculation yields that:

$$\theta^* = \frac{\frac{\Phi(\theta_{xxx})\theta_{xxy}-\Phi(\theta_{xxy})\theta_{xxx}}{\theta_{xxx}-\theta_{xxy}} - \frac{\Phi(\theta_{\varnothing xx})\theta_{\varnothing xy}-\Phi(\theta_{\varnothing xy})\theta_{\varnothing xx}}{\theta_{\varnothing xx}-\theta_{\varnothing xy}}}{\frac{\Phi(\theta_{xxx})-\Phi(\theta_{xxy})}{\theta_{xxx}-\theta_{xxy}} - \frac{\Phi(\theta_{\varnothing xx})-\Phi(\theta_{\varnothing xy})}{\theta_{\varnothing xx}-\theta_{\varnothing xy}}}.$$

Define $\Delta(\alpha) = \frac{\theta_{xxx}+\theta_{xxy}}{2} - \theta^*$. Then, we also have that:

$$\Delta(\alpha) = \Phi(\theta_{xxx}) + \Phi(\theta_{xxy}) - \left(\frac{\Phi(\theta_{\varnothing xx})(\theta_{xxx} + \theta_{xxy} - 2\theta_{\varnothing xy}) - \Phi(\theta_{\varnothing xy})(\theta_{xxx} + \theta_{xxy} - 2\theta_{\varnothing xx})}{\theta_{\varnothing xx} - \theta_{\varnothing xy}}\right).$$

Note that when $\alpha = 0$, because $\theta_{\varnothing xx} = \theta_{xxx}$, and $\theta_{\varnothing xy} = \theta_{xxy}$, $\Delta(0) = 0$. On the other hand:

$$\Delta'(\alpha)|_{\alpha=0} = -\frac{\frac{\partial\theta_{\varnothing xx}}{\partial\alpha}(\Phi'(\theta_{xxx}) - Q) - \frac{\partial\theta_{\varnothing xy}}{\partial\alpha}(Q - \Phi'(\theta_{xxy}))}{\theta_{xxx} - \theta_{xxy}}$$

where $Q = \frac{\Phi(\theta_{xxx})-\Phi(\theta_{xxy})}{\theta_{xxx}-\theta_{xxy}}$. Note that $\Phi'''(a) \geqslant 0$ implies that $\frac{\Phi'(a)+\Phi'(b)}{2} \geqslant \frac{\Phi(a)-\Phi(b)}{a-b}$.[23] By inequalities (A.10) and $\frac{\Phi'(a)+\Phi'(b)}{2} \geqslant \frac{\Phi(a)-\Phi(b)}{a-b}$:

$$\Delta'(\alpha)|_{\alpha=0} \geqslant -\frac{\partial\theta_{\varnothing xx}}{\partial\alpha}\frac{\Phi'(\theta_{xxx}) + \Phi'(\theta_{xxy}) - 2Q}{\theta_{xxx} - \theta_{xxy}} \geqslant 0,$$

which completes the proof. □

[23] To show this, note that by Taylor expansion, $\frac{\Phi(a)-\Phi(b)}{a-b} = \Phi'(a)+\Phi''(a)(b-a)/2+\Phi'''(c)(b-a)^2/6$ for some $c \in [b, a]$. In addition, let $V(a, b) = \frac{\Phi'(a)+\Phi'(b)}{2} - \frac{\Phi(a)-\Phi(b)}{a-b}$. Then, we can show that $\lim_{a\to b} V(a, b) = 0$ and $\frac{\partial V(a,b)}{\partial a} = \Phi'''(c)(a - b)/6$. This concludes the proof.

Using the definition of γ:

$$\frac{\partial \Gamma(a)}{\partial a} = \frac{\theta_{xxx} - \theta_{xxy}}{\theta_{\varnothing xx} - \theta_{\varnothing xy}} \left(\Phi(\theta_{\varnothing xx}) - \Phi(\theta_{\varnothing xy})\right) - \left(\Phi(\theta_{xxx}) - \Phi(\theta_{xxy})\right)$$

Note that if Φ is strictly convex, for each $a > b$, $\frac{\Phi(a)-\Phi(b)}{a-b}$ is increasing in a and b. Then, because $\theta_{xxx} > \theta_{\varnothing xx} > \theta_{xxy} > \theta_{\varnothing xy}$:

$$\frac{\Phi(\theta_{xxx}) - \Phi(\theta_{xxy})}{\theta_{xxx} - \theta_{xxy}} > \frac{\Phi(\theta_{\varnothing xx}) - \Phi(\theta_{\varnothing xy})}{\theta_{\varnothing xx} - \theta_{\varnothing xy}}.$$

Because $\theta_{xxx} - \theta_{xxy} \geqslant \theta_{\varnothing xx} - \theta_{\varnothing xy}$, we have $\frac{\partial \Gamma(a)}{\partial a} < 0$. Then, because $\Gamma(a^*) = 0$, $\Gamma(a) < 0$ for each $a > a^*$. By inequality $\gamma(a) > a$ and Claim 1, for a sufficiently small α, $\mu > \theta^* = \gamma(a^*) > a^*$, and then $\Gamma(\mu) < 0$. Note that by (A.9), $\mu^* = \gamma(\mu)$. Then:

$$H(\alpha, \Phi) = \gamma(\mu)\Phi(\theta_{\varnothing xx}) + (1 - \gamma(\mu))\Phi(\theta_{\varnothing xy}) - (\mu\Phi(\theta_{xxx}) + (1 - \mu)\Phi(\theta_{xxy})) = \Gamma(\mu),$$

which implies that $H(\alpha, \Phi) < 0$. □

A.4. Proofs in Section 5.2

Before presenting the proofs of the propositions, we prepare the following lemma, which characterizes the condition under which the waiting strategy is better than the consistent recommendation. We use this lemma in the proofs of Propositions 3 and 4.

Lemma 4. *There is a concave quadratic function of θ, $h(\theta\colon r, \beta)$ such that for the expert with $\theta_1 = \theta$, the waiting strategy is preferred to consistent recommendations if and only if $h(\theta\colon r, \beta) \geqslant 0$.*

Proof of Lemma 4. Let $\tilde{h}(\theta\colon r, \beta)$ be the difference between the waiting strategy and the consistent recommendation in terms of the expected utilities. Then, the waiting

strategy is preferred to the consistent recommendation if and only if $\tilde{h}(\theta\colon r,\beta) \geqslant 0$. By a calculation, $\tilde{h}(\theta\colon r,\beta) \geqslant 0$ if and only if:

$$\begin{aligned} h(\theta\colon r,\beta) \coloneqq\ & \alpha\theta^2(A(r,\beta)+B(r,\beta)-C(r,\beta)) \\ & + \theta(B(r,\beta)+\alpha C(r,\beta)+(1-\alpha)A(r,\beta)) - A(r,\beta) \geqslant 0, \end{aligned}$$

where $A(r,\beta) \coloneqq \Phi(\theta_{xxy}) - \Phi(\theta_{\varnothing xy})$, $B(r,\beta) \coloneqq \Phi(\theta_{\varnothing xx}) - \Phi(\theta_{xxx})$ and $C(r,\beta) \coloneqq \Phi(\theta_{\varnothing xx}) - \Phi(\theta_{\varnothing xy}) + K$.[24] Because $A+B-C = \Phi(\theta_{xxy}) - \Phi(\theta_{xxx}) - K < 0$, $h(\theta\colon r,\beta)$ is a concave quadratic function. □

Proof of Proposition 3. (i) Consider the waiting strategy. Suppose that $r_1(\theta) = \varnothing$ for each θ and the evaluator imposes the out-of-equilibrium belief that places a probability of 1 on $\theta = \inf D$ if $r_1 \neq \varnothing$. Then, because $\theta_{xxx} = \theta_{xxy} = \inf D$, when the expert recommends a choice in the first period, there is no difference in reputation regardless of which choice the expert recommends in the second period, and the reputation is below that gained when the expert remains silent. Therefore, we need only consider the monetary reward. However, because the waiting strategy recommends the best choice with respect to the second signal, the other strategies are never better than the waiting strategy. Therefore, the waiting strategy is optimal for each θ and, thus, the pair of the strategy and belief constructs an equilibrium.

(ii) Consider the consistent recommendation. Suppose that the expert employs the consistent recommendation irrespective of her ability and that the evaluator has the out-of-equilibrium belief that places a probability of 1 on $\theta = \inf D$ if $r_1 = \varnothing$. Then, because $\theta_{\varnothing xx} = \theta_{\varnothing xy} = \inf D$, $h_\theta(1/2\colon r,\beta) = \alpha((\inf D) - \theta_{xxy}) + (1+\alpha)(\theta_{xxy} - \theta_{xxx}) < 0$.[25] By Lemma 4, h is single-peaked, which implies that $h(\theta\colon r,\beta)$ is decreasing in

[24] When there is no room for confusion, we simplify the notation to drop (r,β).
[25] By an abuse of the notation, we write $h_\theta = \frac{\partial h}{\partial \theta}$.

$\theta \in [1/2, 1]$. In addition, we have that:

$$h(1/2\colon r, \beta) = \frac{1}{4}\alpha(\Phi(\theta_{xxy}) - \Phi(\theta_{xxx}) - K) + \frac{1}{2}(\Phi(\theta_{xxy}) - \Phi(\theta_{xxx}) \\ + \alpha(K + \Phi(\theta_{xxy}) - \Phi(\inf D))) - (\Phi(\theta_{xxy}) - \Phi(\inf D)).$$

Note that $\theta_{xxy} < \theta_{xxx}$ for each $\alpha \in \mathbb{R}_+$. Then, for a small α, $h(1/2 : r, \beta) < 0$ and, thus, $h(\theta : r, \beta) < 0$ for each $\theta \in D$. Hence, the consistent recommendation is better than the waiting strategy for each θ. When inequality (A.5) holds, as shown in Proposition 1(ii), the second-stage behavior is optimal, and the consistent recommendation is a PBE. □

Proof of Lemma 1. $\mathbb{E}U(\theta\colon r_1 = \varnothing, \beta) - \mathbb{E}U(\theta\colon r_1 = s_1, \beta)$ is calculated as:

$$\tilde{h}(\theta\colon r, \beta) = \theta(\Phi(\theta_{\varnothing xx}) - \Phi(\theta_{xxx})) + (\theta_2 - \theta)(\Phi(\theta_{\varnothing xx}) - \Phi(\theta_{\varnothing xy}) + K) \\ - (1 - \theta)(\Phi(\theta_{xxy}) - \Phi(\theta_{\varnothing xy})).$$

By differentiation, we have:

$$\tilde{h}_\theta(\theta\colon r, \beta) = \Phi(\theta_{\varnothing xx}) - \Phi(\theta_{\varnothing xy}) - (\Phi(\theta_{xxx}) - \Phi(\theta_{xxy})) \\ -\alpha\frac{(2\theta - 1)(1 + \alpha\theta) + \alpha\theta(1 - \theta)}{(1 + \alpha\theta)^2}(\Phi(\theta_{\varnothing xx}) - \Phi(\theta_{\varnothing xy}) + K)$$

Because $\theta \geqslant 1/2$, the third term must be negative.

Consider a PBE that is given in Proposition 3(i). In this case, because the first term of $\tilde{h}_\theta(\theta : r, \beta)$ is positive, and $\theta_{xxx} = \theta_{xxy} = \inf D$, when α is sufficiently small, $\tilde{h}_\theta(\theta : r, \beta) > 0$ for each $\theta \in D$.

Consider a PBE that is given in Proposition 3(ii). Then, $\theta_{\varnothing xx} = \theta_{\varnothing xy} = \inf D$ and $\theta_{xxx} > \theta_{xxy}$. Therefore, $\tilde{h}_\theta(\theta\colon r, \beta) < 0$ for each $\theta \in D$. □

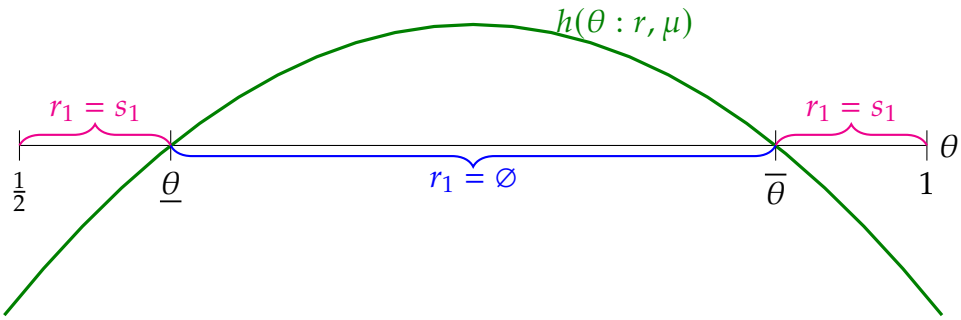

Figure 6: A tripartite equilibrium

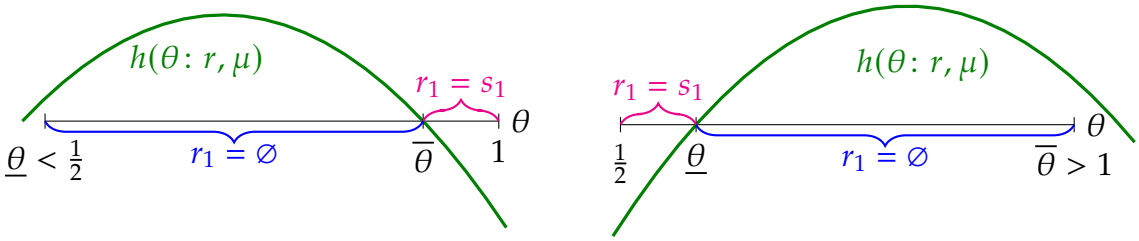

Figure 7: Bipartite equilibria

A.5. Proof of Proposition 4

By Lemma 4, any separating equilibrium has the following properties:

1. There is an interval $D_\varnothing$ such that:

   a) If $\theta \in D_\varnothing \cap D$: $r_1(\theta, s_1) = \varnothing$ and $r_2(\theta, r_1, s_1, s_2) = s_2$.

   b) If $\theta \in D \setminus D_\varnothing$: $r_1(\theta, s_1) = s_1$ and $r_2(\theta, r_1, s_1, s_2) = r_1$ when $r_1 = s_1$.

2. The evaluator believes that if $r_1 = \varnothing$, the expert employs the waiting strategy, whereas if $r_1 \neq \varnothing$, the expert employs the consistent recommendation strategy.

This type of equilibrium is called *tripartite* (Figure 6). When $\underline{\theta} = \inf D_\varnothing < \inf D$ or $\overline{\theta} = \sup D_\varnothing > \sup D$, the tripartite equilibrium degenerates into a bipartite equilibrium (Figure 7). Lemma 5 shows that no bipartite equilibrium exists for any $\alpha \geqslant 0$. Lemma 6 shows that with a sufficiently small $\alpha > 0$, no tripartite equilibrium exists.

Lemma 5. *There is no bipartite equilibrium.*

Lemma 6. *For a sufficiently small $\alpha > 0$, there is no tripartite equilibrium.*

The proofs of these lemmas are provided in Appendix A.6. By Lemmas 5 and 6, when $\alpha > 0$ is sufficiently small, no separating equilibrium associated with ability exists. □

A.6. Proof of lemmas in Appendix A.5

Proof of Lemma 5. Suppose that there is a bipartite equilibrium. Then, either $\overline{\theta} > \sup D$ or $\underline{\theta} < \inf D$ holds.

Suppose that $\overline{\theta} > \sup D$. This implies that $\theta_{xxy} \leqslant \theta_{xxx} < \underline{\theta} < \theta_{\varnothing xy} \leqslant \theta_{\varnothing xx}$. Therefore, $B > 0$ and $A < 0$ and, thus, $h(1\colon r, \beta) > 0$ and $h(1/2\colon r, \beta) > 0$. Because h is a concave quadratic function, $h(\theta\colon r, \beta) > 0$ for each $\theta \in [1/2, 1]$. This shows that the waiting strategy is preferred to the consistent recommendation for each $\theta \in [1/2, 1]$, which is a contradiction.

Suppose that $\underline{\theta} < \inf D$. This implies that $\theta_{xyy} \geqslant \overline{\theta} > \theta_{\varnothing xx}$. Then, for each $\theta < \overline{\theta}$, $h(\theta\colon r, \beta) \geqslant 0$. This condition is rearranged as:

$$\alpha K > \frac{1+\alpha\theta}{\theta(1-\theta)}\left[\theta\Phi(\theta_{xxx}) + (1-\theta)\Phi(\theta_{xxy}) - \frac{1}{1+\alpha\theta}[(1+\alpha)\theta\Phi(\theta_{\varnothing xx}) + (1-\theta)\Phi(\theta_{\varnothing xy})]\right].$$

By $\theta_{xxy} > \theta_{\varnothing xx} > \theta_{\varnothing xy}$:

$$\frac{1+\alpha\theta}{\theta(1-\theta)}\left[\theta\Phi(\theta_{xxx}) + (1-\theta)\Phi(\theta_{xxy}) - \frac{1}{1+\alpha\theta}[(1+\alpha)\theta\Phi(\theta_{\varnothing xx}) + (1-\theta)\Phi(\theta_{\varnothing xy})]\right] > \frac{1+\alpha\theta}{1-\theta}\left(\Phi(\theta_{xxx}) - \Phi(\theta_{xxy})\right).$$

On the other hand, because the consistent recommendation is optimal after $r_1(s_1\colon \theta_1) = s_1$ for each $\theta_1 > \overline{\theta}$, as shown in the proof of Proposition 1(ii), the following inequality

is satisfied:

$$\frac{1}{1+\alpha} > \frac{\Phi(\theta_{xyy}) - \Phi(\theta_{xxy}) + K}{\Phi(\theta_{xxx}) - \Phi(\theta_{xyy}) + K}.$$

This inequality is rearranged as:

$$\alpha K < \Phi(\theta_{xxx}) + (1+\alpha)\Phi(\theta_{xxy}) - (2+\alpha)\Phi(\theta_{xyy}).$$

Because $\theta_{xxx} > \theta_{xyy} > \theta_{xxy}$, we have the following inequality:

$$\Phi(\theta_{xxx}) + (1+\alpha)\Phi(\theta_{xxy}) - (2+\alpha)\Phi(\theta_{xyy}) \leqslant \Phi(\theta_{xxx}) - \Phi(\theta_{xyy}).$$

Therefore, the following inequality holds:

$$\Phi(\theta_{xxx}) - \Phi(\theta_{xxy}) > \alpha K > \frac{1+\alpha\theta}{1-\theta}\left(\Phi(\theta_{xxx}) - \Phi(\theta_{xxy})\right). \tag{A.12}$$

However, because $\Phi(\theta_{xxx}) - \Phi(\theta_{xxy}) > 0$ and $\frac{1+\alpha\theta}{1-\theta} > 1$ for each $\theta \in D$, inequality (A.12) cannot be satisfied. This is a contradiction. □

Proof of Lemma 6. Suppose, by contradiction, that a tripartite equilibrium exists for some small value of α. Without loss of generality, we can assume that there is a sequence of tripartite equilibria with $\alpha \to 0$.

Then, for each $\theta \in D_\varnothing \cap D$, $h(\theta, r, \beta) \geqslant 0$. If $\theta_{xxy} > \theta_{\varnothing xx}$, as in the proof of Lemma 5, we have a contradiction. Thus, we consider the case where $\theta_{\varnothing xx} \geqslant \theta_{xxy}$.

Any tripartite equilibrium, as depicted in Figure 6, requires that $h(1/2, r, \beta) < 0$,

$h_\theta(1/2, r, \beta) > 0$, $h(1, r, \beta) < 0$, and $h_\theta(1, r, \beta) < 0$. Therefore:

$$\alpha X + Y = A + (1 + \alpha)B > 0, \tag{A.13}$$

$$\alpha X + 2Y - 4A < 0, \tag{A.14}$$

$$2\alpha X + Y < 0, \tag{A.15}$$

and $B < 0$, where $X = A + B - C < 0$, $Y = (B + \alpha C + (1 - \alpha)A)$, $A = \Phi(\theta_{xxy}) - \Phi(\theta_{\varnothing xy})$, $B = \Phi(\theta_{\varnothing xx}) - \Phi(\theta_{xxx})$, and $C = \Phi(\theta_{\varnothing xx}) - \Phi(\theta_{\varnothing xy}) + K$. Because $\theta_{xxy} \leqslant \theta_{\varnothing xx}$, $C = \Phi(\theta_{\varnothing xx}) - \Phi(\theta_{\varnothing xy}) + K \geqslant \Phi(\theta_{xxy}) - \Phi(\theta_{\varnothing xy}) = A$. By (A.13) and $B < 0$, $A > 0$ and, thus, $C > 0$. Further, by (A.13) and (A.15), as $\alpha \to 0$, $\lim_{\alpha \to 0} Y = \lim_{\alpha \to 0} A + B = 0$. Otherwise, it has a contradiction.

Note that $h(\theta \colon r, \beta)$ is rearranged as:

$$\alpha X \left(\theta + \frac{Y}{2\alpha X} \right)^2 - \frac{Y^2}{4\alpha X} - A$$

Thus, $h(\theta \colon r, \beta)$ is maximized at $\theta = -\frac{Y}{2\alpha X}$. Because $h(\theta \colon r, \beta) > 0$ at some $\theta \in [1/2, 1]$:

$$-\frac{Y^2}{4\alpha X} - A > 0. \tag{A.16}$$

By rearranging (A.16) and taking a limit $\alpha \to 0$, we have:

$$\lim_{\alpha \to 0} \frac{1}{4} \frac{A + B}{\alpha} \frac{A + B}{A} > \lim_{\alpha \to 0} C. \tag{A.17}$$

On the other hand, by rearranging (A.15) and taking a limit:

$$\lim_{\alpha \to 0} \frac{A + B}{\alpha} < \lim_{\alpha \to 0} C. \tag{A.18}$$

Using (A.18), (A.17) implies that:

$$\lim_{\alpha \to 0} C \frac{1}{4} \frac{A+B}{A} > \lim_{\alpha \to 0} C.$$

This implies that $\lim_{\alpha \to 0} \frac{A+B}{A} > 1$ and, thus, $B > 0$ for a sufficiently small $\alpha > 0$. However, this contradicts that $B < 0$. □

B. Formal analysis in Section 6

Section 6 (informally) discussed the extensions. The following subsections supplement the discussion with formal statements.

B.1. Different accuracy growth rates

This subsection considers different accuracy growth rates, which are assumed to be constant in the basic model. In this subsection, we first assume that the growth rate α is a function of ability θ_1. Then, because in the basic model the consistent recommendation is an equilibrium when:

$$\frac{1}{1+\alpha(\theta_1)} \geqslant \frac{\Phi(\tilde{\theta}^C_{xyy}) - \Phi(\tilde{\theta}^C_{xxy}) + K}{\Phi(\tilde{\theta}^C_{xxx}) - \Phi(\tilde{\theta}^C_{xyx}) + K}, \tag{A.19}$$

where:

$$\tilde{\theta}^C_{xxx} = \frac{m_2}{\mu}, \tilde{\theta}^C_{xxy} = \frac{\mu - m_2}{1-\mu}, \tilde{\theta}^C_{xyx} = \frac{\int \frac{\theta^2(1-\theta)}{1+\alpha(\theta)\theta} f(\theta) d\theta}{\int \frac{\theta(1-\theta)}{1+\alpha(\theta)\theta} f(\theta) d\theta}, \tilde{\theta}^C_{xyy} = \frac{\int \frac{\theta^2(1-\theta)(1+\alpha(\theta))}{1+\alpha(\theta)\theta} f(\theta) d\theta}{\int \frac{\theta(1-\theta)(1+\alpha(\theta))}{1+\alpha(\theta)\theta} f(\theta) d\theta}.$$

Now, we have:

$$\tilde{\theta}^C_{xxx} - \tilde{\theta}^C_{xyx} - (\tilde{\theta}^C_{xyy} - \tilde{\theta}^C_{xxy}) = \frac{1}{\mu(1-\mu)\int \frac{\theta(1-\theta)(1+\alpha(\theta))}{1+\theta\alpha(\theta)} f(\theta)d\theta \int \frac{\theta(1-\theta)}{1+\theta\alpha(\theta)} f(\theta)d\theta}$$
$$\times \left[\int \frac{\theta(1-\theta)(1+\alpha(\theta))}{1+\theta\alpha(\theta)} f(\theta)d\theta \int \frac{\theta(1-\theta)}{1+\theta\alpha(\theta)} \left(\frac{1}{2}M - \mu(1-\mu)\theta\right) f(\theta)d\theta \right.$$
$$\left. + \int \frac{\theta(1-\theta)}{1+\theta\alpha(\theta)} f(\theta)d\theta \int \frac{\theta(1-\theta)(1+\alpha(\theta))}{1+\theta\alpha(\theta)} \left(\frac{1}{2}M - \mu(1-\mu)\theta\right) f(\theta)d\theta \right],$$

where $M = (1-2\mu)m_2 + \mu^2$. Assume that $\frac{1+\alpha(\theta)}{1+\theta\alpha(\theta)}$ and $\frac{1}{1+\theta\alpha(\theta)}$ are weakly decreasing in θ. For example, $\alpha(\theta) = a/\theta^b$ and $\alpha(\theta) = a\theta$ satisfy the assumption when $a \geqslant 0$ and $b \in (0,1)$. As another example, consider the case where $\frac{\theta_2}{1-\theta_2} = \left(\frac{\theta}{1-\theta}\right)^{1+\alpha}$. In this case, the growth rate is represented by $\alpha(\theta) = \left(\frac{\theta}{1-\theta}\right)^{\alpha} - 1$. This $\alpha(\theta)$ also satisfies the assumption. Under this assumption, as in the proof of Lemma 3, we can show that $\tilde{\theta}^C_{xxx} - \tilde{\theta}^C_{xyx} - (\tilde{\theta}^C_{xyy} - \tilde{\theta}^C_{xxy}) > 0$. Then, if $\max_{\theta \in D} \alpha(\theta)$ is sufficiently small and Φ is weakly convex, (A.19) holds for each $\theta \in D$. Therefore, the consistent recommendation remains a PBE.

In another case of the different growth ratio, although α is independent of θ_1, α is distributed by a probability density function g on $[0, \bar{\alpha}]$. The result of this case is similar to the previous one. As in the basic model, the consistent recommendation is an equilibrium when:

$$\frac{1}{1+\alpha} \geqslant \frac{\Phi(\hat{\theta}^C_{xyy}) - \Phi(\hat{\theta}^C_{xxy}) + K}{\Phi(\hat{\theta}^C_{xxx}) - \Phi(\hat{\theta}^C_{xyx}) + K}, \tag{A.20}$$

where:

$$\hat{\theta}^C_{xxx} = \frac{m_2}{\mu}, \hat{\theta}^C_{xxy} = \frac{\mu - m_2}{1-\mu}, \hat{\theta}^C_{xyx} = \frac{\iint \frac{\theta^2(1-\theta)}{1+\alpha\theta} f(\theta)d\theta g(\alpha)d\alpha}{\iint \frac{\theta(1-\theta)}{1+\alpha\theta} f(\theta)d\theta g(\alpha)d\alpha}, \hat{\theta}^C_{xyy} = \frac{\iint \frac{\theta^2(1-\theta)(1+\alpha)}{1+\alpha\theta} f(\theta)d\theta g(\alpha)d\alpha}{\iint \frac{\theta(1-\theta)(1+\alpha)}{1+\alpha\theta} f(\theta)d\theta g(\alpha)d\alpha}.$$

Each $\hat{\theta}^C_{r_1 r_2 \omega}$ takes a similar form as in the previous case. Therefore, we can show that $\hat{\theta}^C_{xxx} - \hat{\theta}^C_{xyx} - (\hat{\theta}^C_{xyy} - \hat{\theta}^C_{xxy}) > 0$. If $\bar{\alpha}$ is sufficiently small, and Φ is weakly convex, the consistent recommendation is again an equilibrium.

B.2. Reputation for the accuracy growth rate

Now, consider the case in which, although θ is fixed, α is unknown to the evaluator, and the expert is concerned about the reputation of α. We denote the probability density function of α by g on $\mathbb{R}_+$.

Under the truthful recommendation, let $\alpha_{r_1 r_2 \omega}$ be the assessment of α. Then:

$$\alpha_{xxx} = \frac{\int \alpha \frac{(1+\alpha)\theta^2}{1+\alpha\theta} g(\alpha) d\alpha}{\int \frac{(1+\alpha)\theta^2}{1+\alpha\theta} g(\alpha) d\alpha} = \frac{\int \alpha \frac{(1+\alpha)}{1+\alpha\theta} g(\alpha) d\alpha}{\int \frac{(1+\alpha)}{1+\alpha\theta} g(\alpha) d\alpha} = \alpha_{xyy}$$
$$\alpha_{xxy} = \frac{\int \alpha \frac{(1-\theta)^2}{1+\alpha\theta} g(\alpha) d\alpha}{\int \frac{(1-\theta)^2}{1+\alpha\theta} g(\alpha) d\alpha} = \frac{\int \alpha \frac{1}{1+\alpha\theta} g(\alpha) d\alpha}{\int \frac{1}{1+\alpha\theta} g(\alpha) d\alpha} = \alpha_{xyx}$$

By observing these calculations, we can find that $\alpha_{xxx} = \alpha_{xyy}$ and $\alpha_{xxy} = \alpha_{xyx}$. Then, reputation is independent of the first-period recommendation. Therefore, this problem is reduced to a single-period model. Hence, the truthful recommendation is optimal.

B.3. Competitive recommendation

This section extends our model to a two-expert setting. There are two experts, i and j. These experts' accuracy growth rates have the same value for α, but different abilities, denoted by θ_i and θ_j. Each expert simultaneously recommends a choice at each opportunity $t \in \{1, 2\}$. Each expert knows her ability, but not the other expert's ability. The evaluator does not know the ability of either expert. For each $k \in \{i, j\}$, let $\theta_{k r_{i1} r_{j1} r_{i2} r_{j2} \omega} = \mathbb{E}_\beta[\theta_k \mid r_{i1}, r_{j1}, r_{i2}, r_{j2}, \omega]$ be the evaluator's assessment of

expert k's ability after the state of the world has been revealed. The term r_{kt} is the recommendation of expert k in period $t \in \{1, 2\}$. Expert k is only concerned about $\theta_{kr_{i1}r_{j1}r_{i2}r_{j2}\omega}$. Let μ_k be the mean and m_{2k} be the second-order moment of expert k's ability.

Now, we investigate the experts' incentives.

Case 1. $r_{i1} = r_{j1} = x$. Consider expert j's decision. Suppose that $s_{j2} = x$. Then, recommending x is dominant. Consider the case where $s_{j2} = y$. By updating j's belief about i's ability, the expectation of i's ability is:

$$\mu_{ixxy} := \frac{\frac{\theta_j(1-\theta_j)(-\alpha m_{2i}+(1+\alpha)\mu_i)}{1+\alpha\theta_j}}{\frac{\theta_j(1-\theta_j)(-\alpha\mu_i+(1+\alpha))}{1+\alpha\theta_j}} = \frac{-\alpha m_{2i} + (1+\alpha)\mu_i}{-\alpha\mu_i + (1+\alpha)} > 1/2,$$

which is independent of θ_j. Then, the condition for recommending y is:

$$\frac{\mu_{ixxy}}{1-\mu_{ixxy}}\frac{1}{1+\alpha} \leqslant \frac{\Phi(\theta_{jxxr_{i2}yy}) - \Phi(\theta_{jxxr_{i2}xy}) + K}{\Phi(\theta_{jxxr_{i2}xx}) - \Phi(\theta_{jxxr_{i2}yx}) + K}.$$

Because i recommends x, x is more likely. This incentive for statistical herding (i.e., term $\mu_{ixxy}/(1-\mu_{ixxy})$, Banerjee 1992; Bikhchandani et al. 1992) strengthens the incentive to make the recommendation consistent.

Case 2. $r_{i1} = y$, $r_{j1} = x$. Consider expert j's decision. Suppose that $s_{j2} = x$. The condition for recommending x is:

$$\frac{1-\mu_{iyxx}}{\mu_{iyxx}}\left(\frac{\theta_j}{1-\theta_j}\right)^2(1+\alpha) \geqslant \frac{\Phi(\theta_{jyxr_{i2}yy}) - \Phi(\theta_{jyxr_{i2}xy}) + K}{\Phi(\theta_{jyxr_{i2}xx}) - \Phi(\theta_{jyxr_{i2}yx}) + K},$$

where $\mu_{iyxx} = \mu_{ixxy}$.

When $s_{j2} = y$, the condition for recommending y is:

$$\frac{1-\mu_{iyxy}}{\mu_{iyxy}}\frac{1}{1+\alpha} \leqslant \frac{\Phi(\theta_{jyxr_{i2}yy}) - \Phi(\theta_{jyxr_{i2}xy}) + K}{\Phi(\theta_{jyxr_{i2}xx}) - \Phi(\theta_{jyxr_{i2}yx}) + K},$$

where $\mu_{iyxy} = \frac{\alpha m_{2i}+\mu_i}{\alpha\mu_i+1}$.

As in the previous case, the incentive for statistical herding (i.e., term $(1-\mu_{ir_is_{j1}s_{j2}})/\mu_{is_{j1}s_{j2}}$) mitigates the incentive to make the recommendations consistent. However, the incentive for statistical herding is mitigated by the reputation concern because the changing opinion may imply that the expert has low ability (*anti*-herding incentive; Avery and Chevalier, 1999, and Levy, 2004).[26] Therefore, the competition effect becomes ambiguous.

[26]In contrast, if the expert does not know her ability, this effect does not operate.